

Topological Systems with CP2K

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Abstract

Electronic transport and topology probe information that cannot be inferred from band energies alone. Current operators, inter-state overlaps, symmetry representations, and the spatial structure of a Hamiltonian must be treated consistently. We formulate these quantities in the nonorthogonal Gaussian atomic-orbital representation of CP2K, connecting finite-temperature Kubo transport, Wilson-loop and Chern analysis, crystalline symmetry representations, spectral localizers, and the quadratic pseudospectrum. The overlap metric is retained in operator products, symmetry transformations, and spectral-gap estimates. Real-space and reciprocal-space constructions share Hamiltonian, position, and spin-orbit matrix elements without requiring a fitted Wannier Hamiltonian. Dense reference calculations are complemented by distributed sparse assembly, symmetric-indefinite inertia calculations, and an optional sparse Pfaffian backend. We explain how the finite matrix formulations relate to noncommutative geometry and distinguish local topological protection from an electronic band gap or a mobility gap. Wilson and inversion calculations, native projective little-group and antiunitary corepresentation tests, and serial/MPI operator comparisons provide complementary validation. The implementation scope is stated explicitly: the Kubo path evaluates dissipative DC response and an optional antisymmetric charge response with scalar relaxation. Crystalline analysis includes nonsymmorphic endpoint-to-line compatibility, but a complete space-group EBR classifier and universal symmetry reduction of topological properties remain development targets.

Keywords: CP2K, Kubo transport, noncommutative geometry, spectral localizer, topological quantum chemistry, spin-orbit coupling

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1 Introduction

Electronic-structure theory supplies a Hamiltonian and a Hilbert-space representation; extracting transport or topological information requires additional structure. Two calculations can agree in their eigenvalues while disagreeing in matrix elements of position, current, or a symmetry operator. This distinction is especially important in a localized nonorthogonal basis, where ordinary matrix multiplication is not the representation of operator composition.

The QUICKSTEP module of CP2K combines localized Gaussian orbitals with plane-wave and multigrid representations of densities and electrostatics [1, 2, 3, 4]. This representation provides access to spatially resolved operators without first fitting a tight-binding Hamiltonian. It also creates specific obligations: the AO metric, periodic image indices, Bloch convention, nonlocal pseudopotentials, and spinor ordering must remain consistent across every analysis.

The first-principles transport programme of Kühne, Heske and Prodan [5] supplies an important starting point: thermally disordered structures are treated as configurations of a covariant family of Hamiltonians, and dissipation is included in the finite-temperature response. Bellissard’s operator-algebraic framework connects this viewpoint to topology without assuming exact translation symmetry of each configuration [6, 7]. The present manuscript combines this motivation with reciprocal-space invariants and spatially resolved finite-matrix diagnostics.

The scope is deliberately broader than a band-topology interface. Wilson loops resolve the geometry of an isolated band subspace; crystalline representations resolve symmetry constraints; a localizer tests an energy-position query; a quadratic pseudospectrum produces approximately localized states; and Kubo response determines a transport coefficient at specified temperature and dissipation. None of these outputs can simply be substituted for another.

The companion study on symmetry-reduced k-point sampling and Wannier analysis develops the underlying Bloch reconstruction and full-mesh export. We repeat the operator conventions needed to make the present article self-contained, but keep the detailed topology, transport, and localizer analysis here. In particular, Wannier90 is an optional route to a reduced representation, not a prerequisite for native AO analysis. Sections below distinguish implemented algorithms from the more general theory and from extensions still requiring validation. The materials-discovery applications discussed later are distinguished from our small test and demonstration calculations: the former motivate the scientific use of these methods, while the latter establish specific numerical identities and illustrate their operation.

2 Hamiltonians and operators in a nonorthogonal basis

2.1 Covariant matrices and the physical metric

Let Φ denote the row of basis functions $|\phi_\mu\rangle$, and $|\psi\rangle = \Phi c$. Define

$$S = \Phi^\dagger \Phi, \quad H = \Phi^\dagger \hat{H} \Phi, \quad X_a = \Phi^\dagger \hat{r}_a \Phi. \quad (2.1)$$

The overlap must be positive definite on the retained basis, and the normalization is $c^\dagger S c = 1$. The generalized eigenproblem is

$$H C = S C \varepsilon, \quad C^\dagger S C = I. \quad (2.2)$$

A covariant matrix A acts on coefficient vectors through $S^{-1} A$. Consequently, composition of two *projected* operators is represented by $A S^{-1} B$, and the projected commutator by $A S^{-1} B - B S^{-1} A$. An ordinary product AB is generally incorrect.

For a nonsingular change of basis $\Phi' = \Phi T$,

$$S' = T^\dagger S T, \quad A' = T^\dagger A T, \quad c' = T^{-1} c. \quad (2.3)$$

The expectation $c^\dagger A c$ and the generalized spectrum are unchanged. The orthonormal representation $\tilde{A} = S^{-1/2} A S^{-1/2}$ is useful for proofs and dense reference calculations. It need not be formed in a sparse implementation: products can be evaluated by solving linear systems with S .

A finite-basis projection is not an algebra homomorphism for arbitrary continuum operators. With $\Pi = \Phi S^{-1} \Phi^\dagger$, $\Phi^\dagger \hat{A} \Pi \hat{B} \Phi$ differs in general from $\Phi^\dagger \hat{A} \hat{B} \Phi$. Thus a commutator or a variance of projected positions is an observable of the projected model, not automatically its complete-basis continuum counterpart. This distinction is monitored through basis convergence rather than hidden in a numerical tolerance.

2.2 Finite positions and AO export

For an isolated system, analytic Gaussian first moments provide X_a in the same basis ordering as H and S . Translation of the coordinate origin by x_a gives $X_a - x_a S$, not $X_a - x_a I$. Likewise an energy query uses $H - E S$. These two simple identities are essential for origin and energy-shift invariance.

The `AO_MATRICES` print section can export overlap, scalar KS Hamiltonian, and the three position matrices. They permit an independent finite-Hamiltonian calculation. Activating SOC in a property section does not silently turn the separate scalar KS export into a spinor Hamiltonian export. Model Hamiltonians can use the low-level matrix kernels when the metric, positions, and symmetry structure are supplied consistently; this does not imply automatic adapters for every electronic-structure method.

2.3 Bloch operators and directed overlaps

In the cell Fourier convention,

$$H_{\mu\nu}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} H_{\mu\nu}(\mathbf{R}), \quad S_{\mu\nu}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} S_{\mu\nu}(\mathbf{R}). \quad (2.4)$$

Here $\mathbf{k}$ is Cartesian and includes inverse-length units; fractional reciprocal coordinates instead require 2π in the phase. Image-cell labels cannot be discarded before forming derivatives or inter-k-point connections.

For cell-periodic states and a directed step $\mathbf{b}$, the occupied-state overlap is

$$M(\mathbf{k}, \mathbf{b}) = C^\dagger(\mathbf{k}) O(\mathbf{k} + \mathbf{b}; \mathbf{b}) C(\mathbf{k} + \mathbf{b}), \quad O_{\mu\nu} = \sum_{\mathbf{R}} e^{i(\mathbf{k}+\mathbf{b})\cdot\mathbf{R}} \langle \phi_{\mu 0} | e^{-i\mathbf{b}\cdot\mathbf{r}} | \phi_{\nu \mathbf{R}} \rangle. \quad (2.5)$$

The zero-step limit is $S(\mathbf{k})$, but a finite link is not obtained by replacing O with S , or by simply contracting coefficient vectors. Neighbor lists, reciprocal-boundary phases, and the selected band rank are part of the observable. Wannier90's file and library interfaces can consume such data [8]; native Wilson calculations need no Wannier localization or interpolation.

2.4 Spin-orbit coupling and symmetry

The implementations considered here start from a converged scalar restricted SCF potential and add GTH spin-orbit matrix elements. The localizer and Bloch-Kubo paths use the complete AO spinor space, whereas a second-variational band-topology calculation uses a selected scalar-state space whose size must be converged separately. Neither construction is self-consistent noncollinear SOC DFT.

In spin-major ordering, the overlap is $\mathbb{S} = \text{diag}(S, S)$ and physical time reversal is $\Theta = JK$, where

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \quad \Theta^2 = -I. \quad (2.6)$$

Time-reversal relations must be checked for the Hamiltonian, metric, and relevant operators; the presence of an SOC pseudopotential alone does not establish them. A Cartesian spatial rotation R

acts on spin through an $SU(2)$ lift of $\det(R)R$, with inversion acting trivially on spin and separately on orbital parity. The sign ambiguity of this lift is retained in the double-group multiplication law.

3 Noncommutative geometry and physical interpretation

3.1 Covariance rather than exact order

For a family of configurations $\omega \in \Omega$, a spatial action τ_a is represented by

$$U_a H_\omega U_a^{-1} = H_{\tau_a \omega}. \quad (3.1)$$

This covariance is weaker than $[U_a, H_\omega] = 0$: a displaced thermal configuration need not have the symmetries of its probability measure. Bellissard’s formulation replaces the Brillouin-zone description by an algebra of such covariant observables with a trace per physical volume [6]. The operator-algebraic viewpoint therefore remains meaningful when a single Bloch momentum is not a good quantum number.

For sufficiently local observables, the trace per volume has the schematic realization

$$\mathcal{T}(A) = \lim_{V \rightarrow \infty} \frac{1}{V} \text{Tr}(\chi_V A \chi_V), \quad (3.2)$$

with the appropriate orbital trace and a physical-volume normalization. Ergodicity supplies the conditions under which a typical large configuration and an ensemble average agree. It does not guarantee that a particular small simulation cell has self-averaged.

The first-principles construction must preserve covariance while passing from continuum KS operators to AO matrices [5]. The metric is part of that transformation. In a finite AO space, traces of projected covariant matrices use $\text{Tr}(S^{-1}A)$; traces of products use the intervening metric factors. For local traces the spatial cutoff itself is an operator, not merely an arbitrary subset of coefficient indices.

3.2 Derivations, Fermi projections, and Chern pairings

Choose spatial derivations $\nabla_a A = -i[X_a, A]$ in an orthonormal representation. For the Fermi projection $P = \mathbf{1}_{(-\infty, E_F)}(H)$, a two-dimensional Chern pairing can be written, with a fixed orientation, as

$$C = 2\pi i \mathcal{T}(P[\nabla_1 P, \nabla_2 P]) = -2\pi i \mathcal{T}(P[[X_1, P], [X_2, P]]). \quad (3.3)$$

The trace normalization and oriented coordinates are chosen so that the periodic limit agrees with the Berry-curvature convention used below. Reversing that orientation reverses C . Index theory relates such pairings to stable integers under the required locality and gap or localization assumptions [7]. Equation (3.3) is a theoretical connection, not a claim that the current implementation evaluates a general thermodynamic-limit noncommutative Chern formula.

A spectral gap excludes eigenvalues in an interval. A mobility gap instead concerns localization and transport even when states are present. A localizer gap is the smallest singular value of a composite energy-position operator. These three gaps answer different questions. The localizer is useful precisely because spatially resolved information can remain available where a bulk band classification is not directly applicable [9]. Its numerical invertibility does not by itself prove a mobility-gap theorem for a DFT material.

3.3 Thermal disorder and convergence

An AIMD trajectory can supply configurations for a disorder average. The nuclear sampling temperature, the electronic Fermi temperature, and the dissipative relaxation scale are conceptually distinct. A calculation should record all three rather than infer them from a single input

temperature. Configuration correlations determine how many effectively independent samples have been obtained; a small standard error computed from strongly correlated snapshots is not a substitute for a correlation analysis.

The thermodynamic limit, zero-temperature limit, and vanishing dissipation limit cannot generally be interchanged. At fixed finite volume the spectrum is discrete, so taking the broadening to zero first can give a qualitatively misleading DC response. For a material prediction we therefore require size and sampling convergence at a specified positive dissipation, followed by a physically justified study of its dependence. The present validation examples establish numerical identities at fixed inputs, not a microscopic scattering model or a disorder-converged transport prediction.

4 Finite-temperature Kubo transport

4.1 Liouvillian formulation and dissipation

Let $f(H) = [1 + \exp((H - \mu)/k_B T)]^{-1}$ and define the energy-valued Liouvillian $\mathcal{L}_H(A) = i[H, A]$. In the relaxation-time approximation, a positive energy broadening Γ replaces a more general dissipative superoperator. With $\nabla_a = -i[X_a, \cdot]$, a convenient conductivity convention is

$$\sigma_{ab} = -\frac{e^2}{\hbar} \mathcal{T}[(\nabla_a H)(\Gamma + \mathcal{L}_H)^{-1} \nabla_b f(H)]. \quad (4.1)$$

The real symmetric part gives the dissipative DC response; the real antisymmetric part gives the Hall response. Throughout this section, the first tensor index denotes current and the second electric field. The factor $e > 0$ is the magnitude of the electron charge. This expression states the operator structure behind the finite-volume calculation, including the role of dissipation [5]. A constant Γ is an approximation to scattering; it is not obtained from the SCF convergence tolerance or from a plotted density-of-states width.

Let $a_{nm}^b = \langle n | \nabla_b H | m \rangle = \hbar v_{nm}^b$, and define the nonnegative Fermi divided difference

$$F_{nm} = \begin{cases} -\frac{f(\varepsilon_n) - f(\varepsilon_m)}{\varepsilon_n - \varepsilon_m}, & \varepsilon_n \neq \varepsilon_m, \\ -f'(\varepsilon_n) = \frac{f(\varepsilon_n)[1 - f(\varepsilon_n)]}{k_B T}, & \varepsilon_n = \varepsilon_m. \end{cases} \quad (4.2)$$

A spectral expansion of Eq. (4.1) gives

$$\sigma_{ab}^{\text{diss}} = \frac{e^2}{\hbar V_d} \sum_{\mathbf{k}, s} w_{\mathbf{k}} g_s \sum_{nm} F_{nmks} \frac{\Gamma}{\Gamma^2 + (\varepsilon_{nks} - \varepsilon_{mks})^2} \Re[a_{nmks}^a a_{mnks}^b]. \quad (4.3)$$

Here $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$, V_d is the periodic length, area, or volume, and g_s accounts for spin occupancy without double counting. For explicit SOC spinors, each state has unit maximum occupation. The tensor is projected onto the periodic Cartesian subspace before reporting a reduced-dimensional conductivity.

Equation (4.3) is positive semidefinite for $T, \Gamma > 0$. In particular, its diagonal elements are nonnegative. The units are conductance times length^{2-d}: S m in one dimension, S in two dimensions, and S/m in three dimensions, with the latter optionally printed as S/cm. Vacuum padding must not change a two-dimensional sheet conductivity through an arbitrary three-dimensional volume factor. An isolated finite-system response is not automatically a measurable bulk conductivity.

4.2 Antisymmetric charge response and its sign

Write $\Delta_{nm} = \varepsilon_n - \varepsilon_m$ and $q_{nm}^{ab} = a_{nm}^a a_{mn}^b$. The resolvent acting on the mn element of the drive has denominator $\Gamma - i\Delta_{nm}$. Consequently,

$$\Re \frac{q_{nm}^{ab}}{\Gamma - i\Delta_{nm}} = \frac{\Gamma \Re q_{nm}^{ab} - \Delta_{nm} \Im q_{nm}^{ab}}{\Gamma^2 + \Delta_{nm}^2}. \quad (4.4)$$

The first term gives Eq. (4.3), while the second yields

$$\sigma_{ab}^A = -\frac{e^2}{\hbar V_d} \sum_{\mathbf{k},s} w_{\mathbf{k}} g_s \sum_{nm} F_{nm\mathbf{k}s} \frac{\Delta_{nm\mathbf{k}s}}{\Gamma^2 + \Delta_{nm\mathbf{k}s}^2} \Im q_{nm\mathbf{k}s}^{ab}. \quad (4.5)$$

Hermiticity gives $q_{nm}^{ba} = (q_{nm}^{ab})^*$, hence $\sigma_{ba}^A = -\sigma_{ab}^A$ and $\sigma_{aa}^A = 0$. This contribution does not enter the isotropic trace or Joule heating.

The implementation stores $J_a = -ia^a$, not the Hermitian velocity. Pairing n, m therefore gives the equivalent kernel $2f[\varepsilon_n, \varepsilon_m] \Delta_{nm} \Im[(J_a)_{mn}(J_b)_{nm}]/(\Gamma^2 + \Delta_{nm}^2)$ for $n < m$, where $f[\varepsilon_n, \varepsilon_m] = -F_{nm}$. Retaining both this minus sign and the tensor-index convention is essential. The kernel is checked against an independently assembled complex Liouvillian in a nondiagonal Hamiltonian frame.

For $T, \Gamma > 0$ the finite-temperature divided difference remains bounded at a degeneracy. Exactly degenerate pairs contribute zero to Eq. (4.5); nearly degenerate pairs approach this limit continuously. Unitary rotations within either degenerate eigenspace preserve the summed contraction. No inverse band splitting is needed in the implemented kernel.

In a gapped two-dimensional orthogonal model, take the cold and weak-relaxation limits only after resolving the gap and $\mathbf{k}$ integration. With Berry connection $\mathcal{A}_n = i\langle u_n | \nabla_{\mathbf{k}} u_n \rangle$ and curvature $\Omega_{n,xy} = -2\Im \sum_{m \neq n} q_{nm}^{xy} / \Delta_{nm}^2$, Eq. (4.5) becomes

$$\sigma_{xy}^A = -\frac{e^2}{\hbar} \sum_{n \text{ occupied}} \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \Omega_{n,xy} = -\frac{e^2}{h} C. \quad (4.6)$$

This sign refers to the explicitly stated current/field convention. A finite relaxation scale, partial occupation or an unconverged mesh does not in general give an integer conductance.

The opt-in `HALL_RESPONSE` setting combines Eq. (4.5) with the dissipative tensor for the projected-AO and Bloch operators and separately prints its three independent entries. Under time reversal the charge Hall tensor changes sign. A time-reversal-invariant Hamiltonian therefore has zero net charge Hall response, including after GTH SOC is added to a restricted SCF. Enabling this output does not create magnetic order or an applied field.

4.3 Degenerate levels and intraband response

The equal-energy branch of Eq. (4.2) applies to all matrix elements within a degenerate subspace, not just $n = m$. In a periodic system, the diagonal band velocity can be nonzero and provides an intraband term proportional to $-f'/\Gamma$. Discarding diagonal terms would remove this contribution. Likewise, dividing current matrix elements by $\varepsilon_n - \varepsilon_m$ cannot be used to define a stable position operator in a degenerate block.

A unitary change of basis inside an exactly degenerate subspace rotates the complete current block. Because the scalar kernel is constant on that block, the summed contraction is unchanged. This is why comparisons should involve block traces or projectors rather than phases of individual eigenvectors. Near equal energies, the implementation evaluates the divided difference with a smooth finite-temperature expansion instead of subtracting nearly identical Fermi functions.

4.4 Three current-operator choices

The historical atom-embedding approximation associates each AO with its nuclear center. It provides an economical representation but does not contain all intra-atomic Gaussian position matrix elements. It remains a distinct, explicitly selected approximation.

For isolated systems, the projected-AO alternative uses

$$J_a = HS^{-1}X_a - X_aS^{-1}H, \quad \hbar v_a = iJ_a. \quad (4.7)$$

J_a is anti-Hermitian. In an S -orthonormal eigenbasis it becomes $(J_a)_{nm} = (\varepsilon_n - \varepsilon_m)(X_a^{\text{MO}})_{nm}$. The complex conjugation in a conductivity contraction must be retained when SOC is present. This construction is exact for the projected finite AO model; the difference between projected and continuum commutators remains a basis-convergence question.

For periodic calculations, a bounded Cartesian position in one simulation cell is not a valid replacement for the periodic current. The Bloch path uses derivatives of the image Hamiltonian and overlap together with analytic first-moment integrals. In atomic units its matrix elements are

$$v_{nm}^a(\mathbf{k}) = [C^\dagger \partial_a H C]_{nm} - \varepsilon_n [C^\dagger \partial_a S C]_{nm} + i(\varepsilon_n - \varepsilon_m) [C^\dagger D_a C]_{nm}, \quad (4.8)$$

where ∂_a differentiates Cartesian k_a . The cell-gauge connection obeys

$$D_a - D_a^\dagger = -i \partial_a S. \quad (4.9)$$

These identities make explicit why the derivative of H alone, the canonical momentum, and the full velocity are not interchangeable in a local-orbital calculation [10].

Hermiticity can be checked directly: the anti-Hermitian part from the energy-weighted overlap derivative is cancelled by Eq. (4.9). For $n = m$, Eq. (4.8) reduces to the generalized Hellmann–Feynman derivative $c_n^\dagger (\partial_a H - \varepsilon_n \partial_a S) c_n$. No eigenvector derivative or energy-denominator division is required.

For distinct band energies, the same operator determines the interband position (dipole) matrix elements in atomic units,

$$d_{nm}^a(\mathbf{k}) = -\frac{i v_{nm}^a(\mathbf{k})}{\varepsilon_{n\mathbf{k}} - \varepsilon_{m\mathbf{k}}}, \quad \varepsilon_{n\mathbf{k}} \neq \varepsilon_{m\mathbf{k}}. \quad (4.10)$$

Here d_{nm}^a has units of length and does not include the electron charge. Substitution of Eq. (4.8) gives the Hamiltonian-derivative term, the energy-weighted overlap-derivative term, and $[C^\dagger D_a C]_{nm}$. All three are required in a general nonorthogonal local-orbital basis [10]. The k -point moment calculation constructs these matrices from real-space image-cell integrals and can use the SCF mesh, a separate KPOINTS set, or a KPOINT_SET path in the moments print section. Equation (4.10) is only an interband relation. It neither supplies diagonal positions in a periodic system nor replaces the undivided velocity blocks needed for degenerate and intraband transport.

4.5 Gauge covariance and nonlocal operators

Under a differentiable, nonsingular AO basis change $T(\mathbf{k})$, $H' = T^\dagger H T$, $S' = T^\dagger S T$, and $C' = T^{-1} C$. The connection must transform as

$$D'_a = T^\dagger D_a T - i(\partial_a T)^\dagger S T. \quad (4.11)$$

Substitution into Eq. (4.8) cancels all derivatives of T by using $H C = S C \varepsilon$. The physical velocity is therefore unchanged even under a k -dependent nonunitary AO transformation. This is a stronger test than comparing band energies.

The Hamiltonian derivatives include all retained image contributions, including nonlocal pseudopotential and GTH-SOC terms. The connection term is still required; including a derivative of the SOC Hamiltonian does not make the bare Peierls derivative complete. SCF-potential convergence, scalar or spinor basis convergence, and analysis-mesh convergence must be checked independently.

4.6 Property symmetry and practical scope

The optional Bloch-Kubo symmetry path diagonalizes only representatives of its own analysis mesh. Atom permutations, AO rotations, lattice-image phases, physical spin rotations, and

antiunitary conjugation reconstruct the remaining eigenframes. Each candidate frame is tested through $C^\dagger SC = I$, $C^\dagger HC = \varepsilon$, and target/source current covariance. An unacceptable map is not silently inserted into the conductivity.

For a spatial Cartesian operation R , velocities rotate as a polar vector. Time reversal adds the reversal of velocity as well as complex conjugation. In the anti-Hermitian current convention used internally, the extra factor of i must also be conjugated. The response sums the full tensor over the full mesh; it is not an irreducible scalar-weight substitution for anisotropic currents.

The implementation provides dissipative and optional antisymmetric charge DC conductivity with a prescribed scalar relaxation scale. Nonzero Hall signs are validated in analytic lattice models; the native GPW/GAPW SOC fixtures are time-reversal-invariant zero-Hall controls, not converged magnetic-material benchmarks. Spin Hall currents, self-consistent noncollinear magnetic order, a general frequency-dependent response and microscopic scattering superoperators are not supplied by this extension. Its Bloch matrices and eigenframes are a bounded dense reference path, although eigensystems are distributed over MPI ranks. Fewer diagonalizations do not imply proportional memory or runtime savings when target operators are still explicitly evaluated.

5 Band topology: Wilson loops, Chern numbers, and time reversal

The overlap matrices used for Wannier construction also contain the geometry of an isolated band subspace. Following established hybrid-Wannier and Wilson-loop formulations [11, 12], the native analysis in CP2K evaluates Berry phases, hybrid Wannier charge centers, the $\mathbb{Z}_2$ invariant of a time-reversal-invariant plane, and the first Chern number C_1 . The native analysis evaluates Wilson loops and topological invariants directly from Gaussian-basis Bloch-state overlaps, without requiring Z2Pack or a maximally localized Wannier representation. Z2Pack remains an optional external analysis route and a complementary validation tool. Its adapter supplies the same first-principles overlaps to an external implementation with its own line and surface sampling. Neither route requires a fitted tight-binding Hamiltonian.

5.1 Interband geometry and Berry curvature

The velocity and dipole matrix elements in Eqs. (4.8) and (4.10) also connect the local-orbital representation to band geometry. For a nondegenerate band, the moment implementation evaluates the interband expression

$$\begin{aligned}\Omega_n^{\text{mom},\gamma}(\mathbf{k}) &= 2 \operatorname{Im} \sum_{m \neq n} d_{nm}^\alpha(\mathbf{k}) d_{mn}^\beta(\mathbf{k}) \\ &= 2 \operatorname{Im} \sum_{m \neq n} \frac{v_{nm}^\alpha(\mathbf{k}) v_{mn}^\beta(\mathbf{k})}{(\varepsilon_{n\mathbf{k}} - \varepsilon_{m\mathbf{k}})^2},\end{aligned}\tag{5.1}$$

where (α, β, γ) is a cyclic permutation of (x, y, z) . This equation uses the sign convention of the moment output: at identical orientation, $\Omega_n^{\text{mom},z} = -\Omega_{n,xy}$ for the Berry connection $\mathcal{A}_n = i\langle u_n | \nabla_{\mathbf{k}} u_n \rangle$ used in Eq. (4.6). This conversion must also be applied when comparing Hall signs. Overlap-based phases additionally require matching the surface orientation. With Cartesian reciprocal coordinates, $\Omega_n^{\text{mom},\gamma}$ has units of length squared. The formula reveals why small avoided crossings can produce sharply localized curvature and why band-energy convergence alone does not establish convergence of this observable. The band sum, AO basis, and reciprocal-space sampling require separate convergence checks.

The product in Eq. (5.1) is invariant under independent phase changes of nondegenerate eigenstates. At an exact degeneracy, however, the individual-band assignment is not invariant under rotations within the degenerate subspace, and division by the vanishing energy difference is not defined. An isolated group of bands must instead be treated through its projector or

matrix-valued connection. The Wilson construction below implements this subspace viewpoint without resolving arbitrary eigenvectors within the group. The interband expression describes the implemented finite-basis moment representation and is not by itself a proof of complete-basis curvature or an integer Chern number. Nor does this geometric quantity constitute an implementation of Hall transport: Section 4 separately defines the occupation-, relaxation- and current-dependent DC tensor.

5.2 Directed overlaps and Wilson loops

Consider a closed directed path $\mathbf{k}_0, \dots, \mathbf{k}_L$ with $\mathbf{k}_L = \mathbf{k}_0 + \mathbf{G}$ and links $\mathbf{b}_j = \mathbf{k}_{j+1} - \mathbf{k}_j$. For N_{occ} occupied states, each link is the square matrix in Eq. (2.5). In the Gaussian representation, its AO operator is explicitly

$$O_{\mu\nu}(\mathbf{k} + \mathbf{b}; \mathbf{b}) = \sum_{\mathbf{R}} e^{i(\mathbf{k}+\mathbf{b})\cdot\mathbf{R}} \langle \phi_{\mu\mathbf{0}} | e^{-i\mathbf{b}\cdot\mathbf{r}} | \phi_{\nu\mathbf{R}} \rangle. \quad (5.2)$$

Here $\phi_{\mu\mathbf{R}}(\mathbf{r}) = \phi_{\mu}(\mathbf{r} - \mathbf{R})$ denotes an individual localized orbital, not a Bloch sum. The zero-step limit recovers $S(\mathbf{k})$. At finite displacement, however, this is a directed cross-k operator, not a Hermitian matrix at one k-point. Its construction retains ordered atom pairs, image-cell phases, and the same periodic atomic images as the Hamiltonian. Consequently, neither $C^\dagger(\mathbf{k})C(\mathbf{k} + \mathbf{b})$ nor replacement of the operator by $S(\mathbf{k})$ represents the required overlap. Reciprocal-boundary links retain their integer $\mathbf{G}$ shift, implementing $u_{n,\mathbf{k}+\mathbf{G}}(\mathbf{r}) = e^{-i\mathbf{G}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})$ in a consistent boundary gauge.

At finite sampling the occupied-subspace overlap need not be unitary. Each link is factorized and replaced by its unitary polar factor,

$$M_j = U_j \Sigma_j V_j^\dagger, \quad Q_j = U_j V_j^\dagger, \quad \mathcal{W} = Q_0 Q_1 \cdots Q_{L-1}. \quad (5.3)$$

The smallest singular value of M_j diagnoses loss of continuity or an inadequately resolved subspace. A rank-deficient link is rejected. For a unitary change of occupied-band basis $C_j \mapsto C_j G_j$, the link and closed product transform as

$$Q_j \mapsto G_j^\dagger Q_j G_{j+1}, \quad \mathcal{W} \mapsto G_0^\dagger \mathcal{W} G_0. \quad (5.4)$$

Thus the Wilson spectrum is gauge invariant although individual overlaps are not. With the path and phase convention used here,

$$\lambda_n(\mathcal{W}) = e^{2\pi i \bar{x}_n}, \quad \bar{x}_n = \frac{\arg \lambda_n}{2\pi} \pmod{1}, \quad \gamma = \arg \det \mathcal{W}. \quad (5.5)$$

The $\bar{x}_n$ are the hybrid Wannier charge centers in reduced units, and γ is the total loop phase in this convention. Reversing the loop reverses the eigenphases. The absolute center origin follows the atomic and cell-origin convention, whereas the topological classification does not.

For spinful time-reversal symmetry, loops spanning half a reciprocal plane connect two time-reversal-invariant boundaries. The parity of partner exchange in the center spectrum determines the two-dimensional invariant. The largest-gap construction tracks the center z_ℓ of the largest gap between adjacent centers on each sampled loop and counts crossings between successive loops [11, 12], giving

$$\nu = N_{\text{cross}} \pmod{2}. \quad (5.6)$$

Loop resolution and transverse sampling are refined together. Acceptance uses center displacements, neighboring-loop gap separation, boundary Kramers pairing, and stable parity. The occupied rank is fixed and separated from conduction states throughout the sampled surface.

On a closed full reciprocal-space surface, C_1 is obtained instead from the winding of the determinant Wilson phase. With $\gamma_\ell = \arg \det \mathcal{W}_\ell$ and a periodic transverse index, the discrete estimate in the present orientation convention is

$$C_1^{\text{mesh}} = \frac{1}{2\pi} \sum_{\ell} \text{Arg} \exp[i(\gamma_{\ell+1} - \gamma_\ell)]. \quad (5.7)$$

The phase increments must be resolved under refinement, and the final transverse boundary must be sewn to the initial one. Unlike the time-reversal half-plane construction, this calculation covers a full closed surface and does not require time-reversal symmetry. The native implementation checks closure and stable winding under refinement and retains the unrounded winding alongside the integer candidate.

In CP2K, explicit fractional k-points and directed connections are read from a `.nnkp` file with `KPOINTS_SOURCE NNKP`, or generated internally with `KPOINTS_SOURCE WILSON.WILSON_LOOP` evaluates a loop, `Z2` selects the half-plane analysis, and `CHERN` selects the full-surface C_1 analysis. The converged SCF potential is held fixed while the post-SCF eigenstates are obtained. The integration mesh defining the density and the ordered paths probing its band topology are distinct objects. Spin-orbit coupling (SOC) enters through a second-variational spinor Hamiltonian built from the scalar states and pseudopotential SOC matrix elements. For spinor coefficients the link is

$$M_j = C_{j,\uparrow}^\dagger O_j C_{j+1,\uparrow} + C_{j,\downarrow}^\dagger O_j C_{j+1,\downarrow}, \quad (5.8)$$

and band selection refers to this spinor spectrum.

This application makes the role of symmetry particularly clear. Scalar irreducible-mesh weights cannot replace an ordered Wilson product. A path requires explicit states at its points or their symmetry reconstruction with all orbital rotations, spin transformations, and boundary phases. By default the native explicit-path calculation diagonalizes the fixed-potential problem at every requested point. An optional independent property-symmetry path instead reconstructs equivalent eigenframes, as described below. Both retain the complete directed neighbor connectivity; neither replaces it with irreducible-SCF weights.

5.3 Independent property eigenframe reconstruction

The property graph is not an SCF quadrature. In particular, a Wilson half-plane need not contain a complete crystal-symmetry star, and a reciprocal-boundary point may occur more than once with different edge labels. The property planner therefore deduplicates periodic points only for diagonalization, and relates only points actually present in the requested graph. It does not add missing integration points or alter the closure condition for ordinary explicit SCF meshes.

For an operation taking a representative $\mathbf{k}_r$ to $\mathbf{k}_t$, let $A_g(\mathbf{k}_t)$ denote the cell-gauge AO action, including atom permutation, real-spherical-orbital rotation and the image phase $\exp[-2\pi i \mathbf{k}_t \cdot \mathbf{L}_A]$. Here $\mathbf{k}_t$ and $\mathbf{L}_A$ have fractional reciprocal and integer direct-lattice coordinates. The reconstructed scalar frame is

$$\tilde{C}_t = A_g(\mathbf{k}_t) \mathcal{K}^{a_g} C_r, \quad a_g \in \{0, 1\}, \quad (5.9)$$

where $\mathcal{K}$ is complex conjugation. Before using it, the code checks $\tilde{C}_t^\dagger S_t \tilde{C}_t - I$, the full residual $R_t = H_t \tilde{C}_t - S_t \tilde{C}_t \varepsilon_r$, and $\tilde{C}_t^\dagger R_t$. The last check detects projected energy errors that an AO-scaled residual can underestimate in an ill-conditioned basis. The actual frozen target Hamiltonian is used, not an operator symmetrized to force agreement. Inconsistent orbits are rejected without an unreported target diagonalization.

The scalar target gauge is then known, allowing second-variational SOC eigenvectors to be transported with the corresponding spin rotation and, when appropriate, physical time reversal. The implementation uses the conjugate $SU(2)$ representation consistent with its $(\sigma_x, -\sigma_y, \sigma_z)$ SOC convention. The complete reconstructed SOC frame is checked for normalization and its target Hamiltonian residual before band selection. A truncated scalar window that cuts a degenerate group is rejected; convergence of a complete window remains a separate physical requirement.

`WANNIER90%SYMMETRY T` enables this optional path for `NNKP`, `WILSON`, `TRIM`, and `SYMMETRY` point sources, with `K290` or `SPGLIB` selected by `SYMMETRY_BACKEND`. It is independent of SCF mesh reduction and presently requires a restricted scalar SCF. All cross-k AO integrals and reciprocal-boundary links are still evaluated explicitly. This is eigenproblem reduction, not yet a reduction of link construction. Independent scalar representative eigenproblems are distributed

over equal-sized k-point MPI groups, with possibly unequal numbers of representatives per group. `SYMMETRY_PARALLEL_GROUP_SIZE` selects automatic grouping (-1 , the default), one group using all ranks (0), or a positive group size. Each group reconstructs and checks the scalar target frames belonging to its representatives. Bounded H/S operator snapshots are Fourier-transformed locally, without replicating every dense real-space image. The metric, AO-scaled and projected-Hamiltonian tests above are unchanged. One validated frame per group and round is transferred into the global property MO storage, with idle groups joining the same transfer sequence. Temporary frame storage does not grow with orbit size. If the snapshots are unavailable, an explicitly reported collective validation path retains the same physical checks within the memory bound. The SOC stage distributes complete orbits over the same groups: bounded snapshots of the three real-space SOC components are Fourier-transformed into group-local matrix tiles, projected into the reconstructed scalar basis, and diagonalized only at the representatives. Each group checks its partners against their target SOC Hamiltonians before the spectra and selected spinors are collected. The factor i of the imaginary real-space SOC operator is included before conjugating mirrored Fourier blocks. Only one complete representative spinor frame is retained per group at a time. If a bounded operator snapshot is unavailable, an explicitly reported collective assembly path preserves the same orbit reduction and physical checks, without full-mesh rediagonalization.

Directed NNKP/Wilson connections are evaluated in bounded batches over the same groups. Each group receives the two scalar eigenframes for one edge, constructs the ordered cross-k AO operator from cosine and sine snapshots, and projects it into the selected scalar or SOC band space. This operator is not Hermitian: ordered atom pairs and lattice images are retained without adding conjugate mirror blocks. Repeated source or destination frames are reused within a group. Only one selected-band connection per active group is collected before processing resumes in the original graph order. Thus reciprocal boundary phases and Wilson-product ordering remain unchanged, and the retained link batch does not grow with graph size. Idle groups join the same transfer sequence; ownership checks require exactly one contribution for each edge. A bounded-workspace or snapshot failure is reported and uses the existing collective construction. This is parallel evaluation of the complete link graph, not symmetry reduction of its edges. Automatic TQC compatibility graphs use a separate representation-transport path. Multi-node scaling is not established by the local tests.

Tests compare the physical subspaces and the link law $M_{ij}^{\text{rec}} = G_i^\dagger M_{ij}^{\text{full}} G_j$, where $G_i = (C_i^{\text{full}})^\dagger S_i C_i^{\text{rec}}$ is evaluated using independently integrated Gaussian overlaps. Thus a degenerate gauge rotation is permitted, but a missing subspace component is not. Neon/SOC reduces 40 requested points to 15 scalar and SOC diagonalizations. A symmetrically sampled stanene reference reduces four TRIM to two and retains $\nu = 1$. The latter agrees with the unreduced full spectrum on the 10^{-12} hartree scale in the tested setups. An intentionally symmetry-breaking SCF sampling is instead rejected. On a sampling-converged stanene surface, both backends reduce 37,056 requested points to 12,481 scalar and SOC diagonalizations, retain $\nu = 1$, and reproduce all 193 Wilson spectra to below 2.3×10^{-13} in reduced units. This small, 26-AO benchmark does not become faster overall: validating the transported frames outweighs the saved diagonalization time. These comparisons and their numerical integration controls are detailed in the Supporting Information; the counts are not total-runtime speedups.

The separately archived reference in Figure 1 demonstrates the calculation using small test systems: a periodic neon reference and buckled stanene, a representative quantum spin Hall system [13]. The computed invariants are $\nu = 0$ and $\nu = 1$, respectively. The sampling-converged stanene surface contains 193 loops with 192 points per loop. Native and Z2Pack analysis of the same polar-unitarized overlaps agree in the Wilson eigenvalues to 8×10^{-14} . A separately adaptive Z2Pack-driven calculation also yields $\nu = 1$. Operator-level checks independently test the zero-step metric, reverse-link adjoint identity, and displaced atomic centers. These complementary tests connect the local Gaussian integral representation to the global topology of the occupied bands. Inputs, sampling parameters, and figure data are given in the Supporting Information and accompanying calculation records.

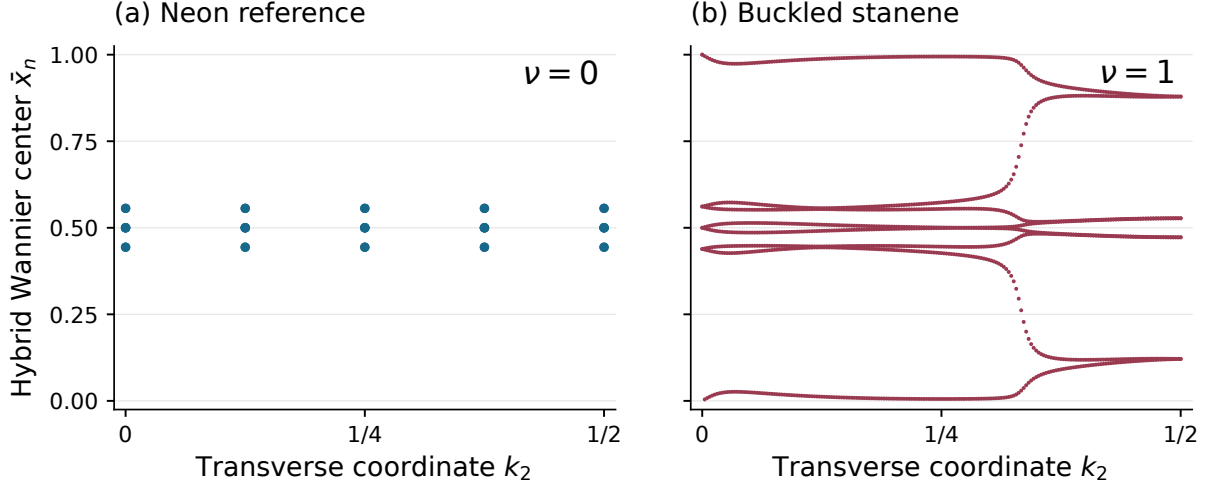


Figure 1: Wilson-loop test and demonstration calculations for (a) a periodic neon reference and (b) buckled stanene. Points are the computed hybrid Wannier centers in reduced units. Degenerate centers coincide. Each loop winds once along the first reciprocal coordinate, while the transverse coordinate spans $0 \leq k_2 \leq 1/2$ at $k_3 = 0$. The neon and stanene surfaces contain five and 193 loops, respectively, and yield $\nu = 0$ and $\nu = 1$. All eight occupied spinor centers are shown.

The historical neon scalar-state truncation cuts a degenerate multiplet. The Supporting Information retains this reference and separately tests complete multiplets and the full scalar basis, distinguishing a stable topological index from thread-independent fine spectral details.

6 Crystalline representations and topological quantum chemistry

6.1 Implemented inversion analysis

An independently versioned native extension evaluates inversion representations at time-reversal-invariant momenta (TRIM). This analysis requires spatial inversion of the crystal, unlike the time-reversal pairing used to reduce a scalar SCF mesh. The TRIM states are obtained at the converged SCF potential independently of the integration mesh, by direct diagonalization or the optional validated reconstruction of Section 5.3, without a Wannier90 execution or a Wannier fit. If $T_{\mathcal{I}}(\mathbf{k})$ is the coefficient-space action of inversion, including atom permutations, orbital parities and lattice phases, its representation in the selected subspace is

$$D_{\mathcal{I}}(\mathbf{k}) = C^{\dagger}(\mathbf{k})S(\mathbf{k})T_{\mathcal{I}}(\mathbf{k})C(\mathbf{k}). \quad (6.1)$$

The overlap metric distinguishes the coefficient transformation $T_{\mathcal{I}}$ from the covariant AO matrix of the inversion operator. Character decomposition gives even and odd multiplicities without assigning gauge-dependent labels to individual vectors within a mixed degenerate eigenspace. For spinful time-reversal symmetry, the number of odd occupied Kramers pairs yields the two-dimensional plane index or the three-dimensional strong and weak Fu–Kane indices [14], together with an inversion z_4 indicator in three dimensions. An exact integer comparison with the analytic inversion-subgroup elementary-band-representation (EBR) signatures also tests membership in their integer lattice and the existence of a nonnegative decomposition [15, 16]. The implementation uses inversion rather than a complete space-group EBR catalogue, and an atomic-compatible signature does not establish trivial topology. Metric, subspace, energy-commutator and time-reversal checks precede classification. Additional parity calculations recover $\nu = 0$ for neon and $\nu = 1$ for stanene. The separately versioned tests are described in the Supporting Information and leave the retained Wilson spectra unchanged.

6.2 Little groups, nonsymmorphic phases, and spin

The extension from inversion to general crystalline symmetry needs more than a list of point rotations. Write a space-group operation as $g = \{W_g | \tau_g\}$ in fractional direct coordinates, so that $x \mapsto W_g x + \tau_g$. For cell matrix A , the Cartesian operation is $R_g = A W_g A^{-1}$. Reciprocal fractional vectors transform with W_g^{-T} , not in general with W_g . An antiunitary operation $g\Theta$ additionally reverses $\mathbf{k}$.

At a fixed fractional $\mathbf{k}$, the little group contains operations mapping $\mathbf{k}$ to itself modulo a reciprocal vector. Choose one representative per coset modulo integer translations of the cell. Composition gives

$$W_{gh} = W_g W_h, \quad L(g, h) = \tau_g + W_g \tau_h - \tau_{gh} \in \mathbb{Z}^3. \quad (6.2)$$

For spinors, the $SU(2)$ lifts additionally give a sign $z(g, h)$. The matrices acting in a fixed Bloch subspace must satisfy

$$D_g^{a_g} D_h = \omega(g, h) D_{gh}, \quad \omega(g, h) = e^{-2\pi i \mathbf{k} \cdot L(g, h)} z(g, h), \quad (6.3)$$

where $a_g = 1$ denotes antiunitarity and ${}^1D_h = \overline{D_h}$. Physical $\Theta^2 = -1$ contributes to the spin factor. Associativity requires the twisted cocycle identity

$$\omega(g, h) \omega(gh, j) = {}^{a_g} \omega(h, j) \omega(g, hj). \quad (6.4)$$

This distinguishes screw/glide operations from a point-group approximation and double-valued representations from scalar ones [17, 19].

A conventional centered cell has distinct cosets with the same rotation and different translations. Collapsing them by rotation alone loses folded translation sectors. Conversely, character data calculated in that conventional quotient cannot be matched blindly to a primitive-cell EBR table. Both the cell transformation and representation convention must be recorded.

6.3 Metric sewing and degeneracies

Let T_g contain the atom/AO action, Bloch-image phases and spin action. The sewing matrix between selected frames is

$$B_g(\mathbf{k}) = C^\dagger(g\mathbf{k}) S(g\mathbf{k}) T_g(\mathbf{k}) {}^{a_g} C(\mathbf{k}). \quad (6.5)$$

For a closed selected subspace it is unitary and reconstructs the transformed frame. Metric normalization, closure, and energy compatibility must be checked before extracting characters. Degeneracy is not resolved by labelling individual eigenvectors: unitary rotations within a degenerate block are physically arbitrary.

For a unitary little group and a complete character table with the same factor system, multiplicities follow from

$$n_\alpha = \frac{1}{|G_{\mathbf{k}}|} \sum_{g \in G_{\mathbf{k}}} \overline{\chi_\alpha(g)} \chi(g). \quad (6.6)$$

The result must be a nonnegative integer, with consistent reconstructed character and total dimension. Antiunitary operations require corepresentation theory, not unmodified use of this character formula. Characters alone also do not determine the sewing of directed Wilson links between different $\mathbf{k}$ -points.

The native implementation constructs a complete projective character table for the supplied unitary little group, rather than importing standard-label tables. Actual GPW/GAPW Gaussian coefficient actions include the atom permutation, AO rotation, cell-image phase and, where requested, the post-SCF GTH spinor action. The selected band sewing matrices are checked against Eq. (6.3) before characters are decomposed. Failed covariance or noninteger multiplicities are rejected, not repaired by rotating eigenvectors or rounding a defective representation.

6.4 Antiunitary corepresentations

Let the little group be $M = H \cup aH$, where H is its unitary subgroup. An irreducible projective character χ_α of H has antiunitary partner

$$\chi_\alpha^a(h) = \frac{\omega(h, a)}{\omega(a, a^{-1}ha)} \overline{\chi_\alpha(a^{-1}ha)}. \quad (6.7)$$

The Wigner indicator is [20]

$$w_\alpha = \frac{1}{|H|} \sum_{b \in aH} \omega(b, b) \chi_\alpha(b^2) \in \{+1, -1, 0\}. \quad (6.8)$$

Type +1 retains one copy of the unitary irrep, type -1 requires two copies, and type 0 pairs two inequivalent partner irreps. The native implementation determines these types and their integer unitary restriction matrix. Odd type- -1 multiplicities or unequal type-0 partner counts invalidate a selected band space. Spin and nonsymmorphic phases enter the same factor system and cannot be tested separately after those phases have been discarded.

These signs are neither topological indices nor antiunitary eigenvalues. Type +1 does not exclude Kramers degeneracy: an already multidimensional spin irrep can contain the required doublet without additional doubling. At a generic centrosymmetric SOC point, in contrast, the identity-only unitary subgroup may be doubled by the combined inversion-time-reversal operation. No canonical antiunitary eigenvector basis is assigned. The electronic-structure path currently constructs the nonmagnetic grey extension of the spatial group. The underlying algebra also accepts supplied magnetic factor systems, but this is not a magnetic space-group search or self-consistent noncollinear SOC calculation.

6.5 Character correspondence and explicit lift conventions

Equal dimensions do not identify irreducible representations. To compare a generated table with named reference characters, let $p(g)$ be a supplied operation isomorphism and let η_g relate the two representation lifts. With $z^{[a_g]} = z$ for a unitary operation and $z^{[a_g]} = \bar{z}$ for an antiunitary operation, consistency requires

$$\omega_t(g, h) = \eta_g \eta_h^{[a_g]} \bar{\eta}_{gh} \omega_r(p(g), p(h)), \quad (6.9)$$

$$\chi_t(g) = \eta_g \chi_r(p(g)), \quad g \in H. \quad (6.10)$$

The operation map must preserve products and antiunitary flags. The phases must be supplied from the actual spatial, Bloch and spin conventions: solving a factor-system coboundary alone cannot fix names, since an additional one-dimensional character can preserve the factors while permuting equal-dimensional irreps. In particular, the spin lifts returned by the factor construction are reused; independently recomputing a lift at a quaternion branch tie can choose the other $SU(2)$ representative through rounding. This includes some cubic rotations as well as π rotations. The Gaussian SOC sewing path uses these same stored lifts, not a second Cartesian-to-spin conversion. For an antiunitary operation the stored spin factor already contains $J = i\sigma_y$; this factor is removed before the frame transformation applies JK , so that physical time reversal is included exactly once. The conjugate spin convention of the CP2K SOC Hamiltonian is retained.

Orthogonality alone is also insufficient to certify a reference table. Let L_g denote the left regular action of the twisted unitary group algebra, $d_\alpha = \chi_\alpha(e)$, and $\nu_\alpha = |H|^{-1} \sum_{g \in H} |\chi_\alpha(g)|^2$. The native matcher constructs

$$P_\alpha = \frac{d_\alpha}{\nu_\alpha |H|} \sum_{g \in H} \overline{\chi_\alpha(g)} L_g \quad (6.11)$$

and checks Hermiticity, idempotence and commutation with every L_g , in addition to character orthogonality and $\sum_\alpha d_\alpha^2 / \nu_\alpha = |H|$. For ordinary unitary irreps $\nu_\alpha = 1$; paired and doubled

Wigner restrictions have norms 2 and 4, respectively. These tests certify the unitary restrictions, not their extension to antiunitary corepresentations, which is checked separately as described above. Only a unique bijection of the complete character columns is accepted. The reference interface does not by itself standardize an arbitrary input cell, name a complete EBR catalogue, or establish global band connectivity.

6.6 Nonsymmorphic band compatibility on explicit segments

For a directed connection from $\mathbf{k}_a$ to $\mathbf{k}_b + \mathbf{G}$, define $\mathbf{d} = \mathbf{k}_b + \mathbf{G} - \mathbf{k}_a$. A segment symmetry must satisfy

$$\sigma_g W_g^T \mathbf{d} = \mathbf{d}, \quad \sigma_g = (-1)^{a_g}, \quad (6.12)$$

within the fractional-coordinate tolerance, not merely modulo a reciprocal vector. Intersection of the endpoint little groups is insufficient: ordinary time reversal can fix two TRIM endpoints without preserving any interior point of their connecting segment.

To compare representations at the endpoints, remove the continuously varying translational part using

$$\eta_g(\mathbf{k}) = e^{2\pi i \mathbf{k} \cdot \boldsymbol{\tau}_g}, \quad \bar{\omega}(g, h) = \eta_g^{a_g} \eta_h \eta_{gh}^* \omega(g, h). \quad (6.13)$$

The conjugation for an antiunitary first factor is essential. Both endpoints must produce the same rephased factor on the segment subgroup. Its unitary projective characters provide a common basis for integer endpoint-to-line branching matrices B_a, B_b . With selected endpoint multiplicities m_a, m_b , compatibility requires

$$B_a m_a = B_b m_b. \quad (6.14)$$

The destination rephasing uses $\mathbf{k}_b + \mathbf{G}$, not just $\mathbf{k}_b$. For example, a screw eigenvalue branch may exchange with another after one reciprocal period. Ignoring $\mathbf{G}$ would erase this monodromy. Integer changes of coset representatives cancel between the characters and the rephasing.

Endpoint corepresentations are restricted as whole representations. If R_a and R_ℓ give the unitary restrictions of endpoint and line corepresentations, their branching C_a satisfies

$$R_\ell C_a = B_a R_a. \quad (6.15)$$

If the line has no antiunitary stabilizer, $B_a R_a$ instead describes the branching into unitary line irreps. This retains endpoint partner constraints even when time reversal is absent on the interior. A spinful glide model illustrates the distinction: opposite glide partners at Gamma become same-branch pairs at the boundary, preventing an isolated two-band connection but allowing a four-band one. For valid corepresentations, unitary and corepresentation line-count compatibility are equivalent; the latter resolves the partner structure rather than introducing a new invariant.

These operations are implemented on explicit NNKP or generated Wilson links through `LITTLE_GROUP_COMPATIBILITY`, which implies `LITTLE_GROUP_IRREPS`. Compatible endpoints do not prove a gap between sampled points, a trivial Bloch bundle, or an atomic-band decomposition. Automatic complete Brillouin-zone connectivity and standard crystallographic irrep labels remain separate work.

The dense Gaussian action is evaluated once per operation across the complete property MPI environment; rows of the sewing-product check are partitioned in the same way. Directed segment restrictions are computed in bounded MPI batches, retaining at most one owned and one received edge per rank. Collective ownership and failure checks precede ordered delivery, including reciprocal-image phases and optional antiunitary endpoint data. This distributes arithmetic, not all storage: AO metrics, selected frames and small representation tables remain replicated. Character tables and the atomic-signature integer lattices described below also remain replicated.

6.7 Site-induced atomic band representations

The native induction kernel starts from a supplied fractional site q and constructs its orbit $q_j = t_j q$ modulo integer translations. Representatives t_j are adjusted by lattice translations to reach the chosen positions exactly. Localized orbitals carry irreps of the site stabilizer; an antiunitary stabilizer instead requires the complete corepresentations described above. Their unitary restriction characters are denoted by χ_α .

For a unitary little-group operation g that fixes site j modulo translations, set

$$L_{gj} = W_g q_j + \tau_g - q_j \in \mathbb{Z}^3, \quad h_j = t_j^{-1} \{I| - L_{gj}\} g t_j. \quad (6.16)$$

The induced character is

$$\chi_\alpha^\uparrow(g; \mathbf{k}) = \sum_{j: gq_j = q_j \bmod \mathbb{Z}^3} e^{-2\pi i \mathbf{k} \cdot L_{gj}} \frac{z(g, t_j)}{z(t_j, h_j)} a_{t_j} \chi_\alpha(h_j). \quad (6.17)$$

This is the Bloch form of site induction [38], with the spin factor in the same convention as Eq. (6.3). Antiunitary orbit representatives conjugate the onsite character even though g itself is unitary. Omitting this conjugation loses complex orbital partners.

The implementation retains conventional-cell centering cosets and keeps each local representation column fixed across $\mathbf{k}$ points. Its band rank is the orbit multiplicity times the local dimension. No full band diagonalization or orbital gauge choice is needed to generate these atomic reference characters. Validation compares their little-group decompositions and segment restrictions with independent Cartesian s/p/d orbital and spin traces. Site induction alone neither enumerates all Wyckoff positions nor establishes that an induced representation is elementary.

6.8 Generating site-symmetry families

The possible atomic reference positions need not be occupied by atoms in the input structure. A separate geometric kernel obtains them from the complete supplied space group. For each operation it solves

$$(W_g - I)q = -\tau_g \pmod{\mathbb{Z}^3} \quad (6.18)$$

and closes the resulting connected fixed sets under nonempty intersection. Saturating the integer row lattice yields primitive normals B for each connected affine torus,

$$Bq = c \pmod{\mathbb{Z}^r}, \quad d = 3 - r. \quad (6.19)$$

Distinct components are retained: for example, $2q_x = 0 \pmod{1}$ contains both $q_x = 0$ and $q_x = 1/2$. Integer right-hand sides are enumerated from the bounds of the unit cube rather than by sampling a coordinate mesh. Subsequent space-group equivalence identifies the families, and a generic representative is checked against its entire site stabilizer.

Closure containment supplies specialization relations between families. A site group is maximal when its family has no higher-symmetry specialization. This provides the geometric input to site induction, but does not replace the representation-level elementarity tests needed for an EBR catalogue. The construction retains the supplied cell and origin conventions and does not assign standard Wyckoff letters.

A separate native matching layer can attach an independently supplied affine reference catalogue to these families. For a reference $q = q_0 + D\alpha$ in the same cell and origin convention, a group operation must satisfy

$$BW_g D = 0, \quad B(W_g q_0 + \tau_g) - c \in \mathbb{Z}^r, \quad \text{rank } D = 3 - r. \quad (6.20)$$

The tangent identity uses bounded integer arithmetic. Orbit multiplicities must also agree, and the complete correspondence must be bijective. Checking the whole affine family avoids treating a

nongeneric reference origin as a generic site. Independent generators and Wyckoff coordinates provide a convention-aware validation of this layer; standard letters remain supplied metadata rather than labels inferred by the Gaussian analysis. This geometric correspondence does not identify onsite irreps with equal dimension.

6.9 Real-space specialization and composite witnesses

Matching little-group labels at sampled momenta is weaker than equivalence of band representations. A stronger constructive test uses a symmetry-preserving displacement of Wannier centers and transitivity of induction [39]. Let q lie in a generic site family with stabilizer H , and let q' be a higher-symmetry endpoint with stabilizer H' . If the lifted path from q to q' is fixed by H , then induction of an onsite representation σ gives

$$\sigma \uparrow H' = \bigoplus_{\beta} B_{\beta\sigma} \rho_{\beta}, \quad \text{BR}_{q,\sigma} \simeq \bigoplus_{\beta} B_{\beta\sigma} \text{BR}_{q',\beta}. \quad (6.21)$$

Unlike a fit of reciprocal signatures, this supplies a real-space equivalence witness. A maximal-site representation can still be composite if it is equivalent through such paths to a direct sum at another site; maximal site symmetry alone is insufficient.

The native implementation solves for an original operation g and integer lift L such that $y = W_g q' + \tau_g + L$ lies in the source's lifted affine closure. Every source stabilizer is checked against

$$(W_h - I)(q - y) = 0, \quad h \in H, \quad (6.22)$$

without reduction modulo lattice vectors. The destination stabilizer is represented as a finite group fixing y exactly, using translations $y - W_h y$. Its orbit of q gives the finite induced representation. Physical spin factors and antiunitary operations are retained. Transport to a different destination orbit image preserves the original seed's representation-column labels.

Decomposition uses complete onsite irreps or corepresentations. For the latter, unitary restrictions are orthogonal but need not have unit character norm. The implementation therefore divides each character inner product by the reference norm and checks the full reconstructed character, integer multiplicities and band dimension. Singleton inductions define equivalence classes; multi-summand inductions provide explicit composite witnesses. Recursive expansion terminates because every summand has strictly smaller positive whole-band dimension. For the resulting nonnegative matrix E , independently generated reciprocal signatures and ranks obey

$$AE = A, \quad N^T E = N^T. \quad (6.23)$$

The output retains lifted endpoints, branching matrices, equivalent representatives and decomposition provenance. It does not identify columns solely from equality in A . An unreduced generator is not, by itself, a certificate of canonical EBR identity or a verdict on the topology of the actual electronic bands.

6.10 Reciprocal fixed families and primitive cycles

The same affine-torus machinery also constructs reciprocal-space symmetry families, consistent with the relation between wave-vector types and reciprocal-space Wyckoff geometry [18]. For an original operation with antiunitary flag a_g , the effective fractional reciprocal action is

$$K_g = (-1)^{a_g} W_g^{-T}, \quad (K_g - I)\mathbf{k} = 0 \pmod{\mathbb{Z}^3}. \quad (6.24)$$

Identical K_g are identified only for geometric enumeration. The full original stabilizer is restored at each generic seed: centering cosets, nonsymmorphic translations and a possible antiunitary PT stabilizer still enter the representation factors. Translation-free reciprocal geometry does not justify discarding those operations.

Each connected closure has saturated normals N and equation $N\mathbf{k} = c \pmod{\mathbb{Z}^r}$. A Smith certificate $UNV = D$ gives a primitive basis $V_{:,r+1:3}$ of its integer tangent lattice. These integer vectors generate torus cycles; an arbitrary floating-point nullspace need not generate the full lattice. For each specialization, an original operation g and integer shift G place its seed in the parent's lifted closure,

$$N_{\text{parent}}(K_g \mathbf{k}_{\text{child}} + G - \mathbf{k}_{\text{parent}}) = 0. \quad (6.25)$$

The code checks closure containment for every child tangent direction and solves the lift as an exact integer equation. It returns explicit points and directed segments, including self-edges with nonzero reciprocal shifts for primitive cycles. Symmetry-related child images are evaluated as additional points rather than matched through assumed irrep labels.

The input mode `KPOINTS_SOURCE SYMMETRY` connects this geometry directly to the Gaussian post-SCF analysis. It obtains eigenstates of the frozen Hamiltonian at the generated points and enables both little-group and segment-compatibility checks, without changing the SCF density or mesh. Optional property symmetry reduces the independent eigenproblems, but retains all points and directed compatibility connections. Uniform interior sampling is controlled by `LITTLE_GROUP_PATH_POINTS`, including the endpoints. If successive lifted samples are $\mathbf{k}_j + G_j$, the stored subedge uses $G_{j+1} - G_j$, not just the target offset. Thus wrapping and deduplicating the sampled points preserve the full directed path and its reciprocal winding, including nonsymmorphic monodromy.

At every retained sample, the selected bands must be separated from computed excluded bands and pass the metric, subspace, Hamiltonian and sewing-law tests. Optional `REQUIRE_GLOBAL_GAP` additionally requires a positive sampled indirect gap for a lowest-band prefix. The name does not imply a rigorous gap certificate between samples; sampling convergence and independent BZ-wide isolation remain necessary. The output retains the geometry, eigenvalues, representations and directed restrictions. These paths are not a Wannier90 neighbour mesh or Wilson surface, so no overlap file is synthesized for them.

Removing higher-symmetry subsets can disconnect a fixed torus, so connected closures are not a complete decomposition into open cells. Their incidence graph and primitive cycles do not by themselves establish complete equivariant band connectivity, primitive-cell equivalence or EBR elementarity.

6.11 Atomic signature matrices on explicit reciprocal sets

Combining the generated sites with the induction kernel gives a native integer signature matrix for a supplied reciprocal set $\mathcal{K}$,

$$A_{(\mathbf{k},\beta),(q,\alpha)} = n_\beta \left[\chi_{q,\alpha}^\uparrow(\cdot; \mathbf{k}) \right], \quad \mathbf{k} \in \mathcal{K}. \quad (6.26)$$

The column (q, α) retains one Wyckoff-family representative and one onsite irrep or corepresentation at every point. Rows use unitary irreps unless the little group has an antiunitary coset; in that case they count complete corepresentations after checking the required unitary partner multiplicities. Consequently,

$$\sum_{\beta} d_{\mathbf{k}\beta} A_{(\mathbf{k},\beta),(q,\alpha)} = N_{q\alpha} \quad (6.27)$$

is the same physical band rank for every $\mathbf{k}$. Counts are not generally numbers of Kramers pairs: corepresentation dimensions and pairing types are retained explicitly.

The implementation preserves all sites, including nonmaximal ones, and does not identify columns merely because their sampled signatures coincide. An incoming unitary character vector can be decomposed in the same row basis after matching its operation indices, provided it uses the same spin/Bloch convention. This algebraic check does not replace Hamiltonian covariance or selected-subspace closure for actual Gaussian states. The matrix remains a redundant atomic reference for the explicit set $\mathcal{K}$, not an automatically complete EBR matrix: reciprocal-stratum

completeness and representation elementarity remain separate from the integer decomposition of this supplied matrix.

6.12 Exact signed and nonnegative signature decompositions

For an integer signature matrix $A \in \mathbb{Z}^{m \times n}$, the native integer kernel constructs a Smith form [38],

$$UAV = D, \quad D = \text{diag}(d_1, \dots, d_r, 0, \dots), \quad d_i > 0, \quad d_i \mid d_{i+1}, \quad (6.28)$$

where U and V are unimodular integer matrices. With $c = Ub$, the equation $Ax = b$ has an integer solution precisely when

$$c_i = 0 \quad (i > r), \quad c_i \equiv 0 \pmod{d_i} \quad (i \leq r). \quad (6.29)$$

A witness is obtained as $x = Vy$, with $y_i = c_i/d_i$ for $i \leq r$ and zero free components. The implementation checks both $UAV = D$ and $Ax = b$ using bounded 64-bit integer arithmetic. Rank decisions do not use floating-point thresholds. Growth beyond the arithmetic bound or exhaustion of the factorization work limit returns an unresolved status, not a negative membership decision.

For the nonnegative signature matrices used here, the stronger condition $Ax = b$, $x \in \mathbb{Z}_{\geq 0}^n$, is tested separately. At a search node with residual $b' \geq 0$, a nonzero column j has the finite bound

$$0 \leq x_j \leq \min_{i: A_{ij} > 0} \left\lfloor \frac{b'_i}{A_{ij}} \right\rfloor. \quad (6.30)$$

Row-wise gcds of the remaining columns and their zero supports prune the complete enumeration; zero columns can be set to zero. Every accepted witness is checked exactly. A caller-specified node budget bounds the potentially exponential search. An exhausted budget is explicitly undecided and must not be interpreted as absence of an atomic decomposition.

These are decisions about the supplied columns and sampled rows. The Smith divisors of an incomplete reciprocal sample are not, by themselves, the space group's symmetry-indicator group. Nonnegative membership likewise does not prove triviality of a Bloch bundle. The integer routines do not add an automatic general-classification input mode or replace spectral isolation and compatibility checks.

6.13 Joint compatibility lattice and atomic quotient

The separate edge tests can be assembled into one integer system. Let $n \in \mathbb{Z}^N$ contain the point-specific irrep or full corepresentation multiplicities, and let P_a select the block at point a . For an edge $e : a \rightarrow b + G_e$, the unitary line restrictions give a block row

$$C_e = B_a R_a P_a - B_b R_b P_b, \quad Cn = 0, \quad (6.31)$$

where R_a is the endpoint corepresentation restriction, or the identity at a purely unitary point. The line corepresentation restriction is injective: its disjoint unitary supports and prescribed pairing or doubling uniquely determine valid line counts. Thus these unitary equations also enforce equality of the full line corepresentations. When $a = b$, both terms act on the same block, but their rephasing still uses G_e . A screw-loop relation must not be discarded as an edge connecting a point to itself.

Symmetry-equivalent points must not carry independent labels. Let s map k_a to $k_b + G$, and for a destination little-group operation h set $h_a = s^{-1}hs$. The actual coset representatives obey $hs = \{I|L\}sh_a$, with

$$L = \tau_h + W_h \tau_s - \tau_s - W_s \tau_{h_a} \in \mathbb{Z}^3. \quad (6.32)$$

Writing z for the spin factor at Gamma, including physical time-reversal squaring, the transported unitary character is

$$\chi_b(h) = \frac{z(h, s)}{z(s, h_a)} e^{-2\pi i k_b \cdot L} \begin{cases} \chi_a(h_a), & s \text{ unitary,} \\ \chi_a(h_a)^*, & s \text{ antiunitary.} \end{cases} \quad (6.33)$$

Independent integer changes of endpoint representatives therefore retain the necessary Bloch phase. We check conjugation of the complete product table and the twisted factor law, including antiunitary little-group operations. Character decomposition must give a bijective unitary-irrep map. Restriction to the unitary subgroup then determines the full corepresentation map M_s , which must preserve dimensions. A spanning set of star relations adds $M_s P_a - P_b$ to C . Edge winding and star identification remain distinct constraints.

For the supplied graph $\mathcal{G}$, define

$$\mathcal{B}_G = \ker_{\mathbb{Z}} C, \quad \mathcal{X}_G = \mathcal{B}_G / \text{im}_{\mathbb{Z}} A. \quad (6.34)$$

This follows the compatible-representation/atomic-lattice construction of symmetry indicators [16], but explicitly retains the scope of the supplied graph. A Smith factorization $U_C C V_C = D_C$ of rank s supplies the saturated integer kernel $K = (V_C)_{:,s+1:N}$. Saturation matters: rescaling rational nullspace vectors separately need not produce the entire integer kernel. Every atomic column must admit an exact solution $A = KY$; failure invalidates the reference/compatibility construction.

Writing $q = N - s$, the Smith form $U_Y Y V_Y = D_Y$ of rank r gives

$$\mathcal{X}_G \simeq \mathbb{Z}^{q-r} \oplus \bigoplus_{i=1}^r \mathbb{Z}_{d_i}. \quad (6.35)$$

Factors with $d_i = 1$ are trivial. For a compatible Gaussian signature $b = Kz$, its class is $U_Y z$ modulo the nonzero divisors, with the remaining integer coordinates retained. The class vanishes precisely when a signed atomic decomposition exists. Nonnegative decomposition is a stronger, separate question. An incompatible signature has no quotient class; it is not assigned a nonzero topological index.

The native implementation checks $CK = 0$ and $KY = A$ and exports the matrices, Smith row map, factors and target residues for independent verification. It removes duplicate homogeneous equations without replacing the integer kernel by a floating-point nullspace. Resource or coefficient-growth limits remain unresolved outcomes. These calculations are replicated on MPI ranks, not distributed sparse integer factorizations.

Star relations between supplied points are now included. Missing points or global compatibility relations can leave free factors in Eq. (6.35). They are diagnostics of the supplied lattices, not additional physical topological invariants. Conversely, vanishing free rank only means that the atomic columns span the compatible space rationally at this sample. It does not certify omitted congruence constraints, complete reciprocal coverage, EBR completeness or spectral isolation throughout the Brillouin zone.

6.14 Connecting atomic references to Gaussian bands

The optional `ATOMIC_SIGNATURES` analysis connects the native atomic reference and integer kernels to the validated Gaussian-band representations. It uses explicit NNKP, Wilson or TRIM points, or the generated SYMMETRY points, and implies the little-group analysis. Each selected band space must be separated from computed excluded bands at the sampled point. Metric covariance, subspace closure, Hamiltonian commutation and the projective sewing law are checked before comparison. This supports scalar GPW/GAPW and the existing post-SCF GTH-SOC treatment; physical time reversal requires a restricted SCF. It does not establish a gap between sampled points.

The band and reference operation lists are matched bijectively by their fractional rotations, antiunitary flags and translations modulo the input lattice, not by their numerical list indices. If the reference representative is $g' = \{I|L_g\}g$, its character changes by

$$p_g = \exp(-2\pi i \mathbf{k} \cdot L_g), \quad \chi'(g') = p_g \chi(g). \quad (6.36)$$

The corresponding factor transformation must also agree,

$$\omega'(g, h) = \frac{p_g^{a_g} p_h}{p_{gh}} \omega(g, h), \quad (6.37)$$

with conjugation when g is antiunitary. After this exact convention conversion, characters are decomposed into the reference row basis. Reordered local irrep labels do not affect the result. Different band ranks, missing operations or inconsistent factors are rejected rather than repaired by fitted gauges.

The output retains the reference operations, characters and corepresentation restrictions, the integer matrix and target, and the site/local-representation identity of each column. Available signed and nonnegative witnesses accompany the membership decisions. The nonnegative search budget is controlled by `ATOMIC_SIGNATURE_MAX_NODES`; exhaustion is reported as unresolved. Invalid data or failed witness verification abort instead of producing a physical classification. Segment compatibility can be requested separately for NNKP/Wilson connections and is automatic for SYMMETRY paths. With atomic signatures enabled, these connections also define the joint lattice of Eq. (6.34). The comparison does not automatically identify complete reciprocal strata, standard EBR labels or the topology of an isolated Bloch bundle.

6.15 Band representations and limits of indicators

Elementary band representations (EBRs) provide the indecomposable building blocks of atomic band representations. Restricting them to little groups yields a vector of irrep multiplicities. Compatibility relations constrain the vectors that can connect along high-symmetry lines. A candidate symmetry vector n can be compared with an integer matrix of EBR signatures,

$$Eq = n. \quad (6.38)$$

Integer solvability, nonnegative solvability, and compatibility constraints are different questions. A symmetry indicator is a quotient of compatible representation data by atomic data; it can miss topology invisible to those symmetry eigenvalues [15, 16]. An atomic-compatible signature is not proof of a trivial Bloch bundle. A signed-only decomposition is not, without additional information, a complete diagnosis of fragile topology.

The inversion-subgroup classifier retains its separate analytic signatures and named indicators. The general native little-group/corepresentation, explicit-segment, Wyckoff-family and site-induction machinery now also tests actual band signatures against the generated atomic columns, but does not supply a complete all-space-group classification. Conventional-to-primitive matching, standard labels, complete connectivity and a validated catalogue with elementarity checks are still required. Algebra tests over all ordinary space groups must not be presented as a completed classification of all their materials.

7 Parameter-space topology and phasons

A separate native postprocessor, `topology_phasons`, extends the overlap construction to closed two- or four-parameter families, including periodic atomic displacements (phasons) combined with reciprocal coordinates. It reads `STATE_EXPORT` snapshots and evaluates $M_{ab} = C_a^\dagger O_{ab} C_b$, where O_{ab} includes the Gaussian centers of both geometries, periodic images, and Bloch and boundary-sewing phases. Polar-unitary links give C_1 from determinant plaquette phases [23] and a four-parameter C_2 estimate from full non-Abelian plaquette logarithms [24]. Here C_2 denotes the second Chern character pairing, not generally the second Chern class. The postprocessor requires neither Python, Z2Pack, nor Wannier90 at runtime. It operates on fixed-cell, fixed-rank families with explicit physical endpoint identifications. The Supporting Information gives its separate development version, discretization, and model and Gaussian-snapshot tests. These

tests complement the material examples below without changing their Hamiltonians, sampling, or results.

For a discrete parameter family, let $Q_\mu(x)$ denote the polar link from x to $x + \hat{\mu}$. An oriented plaquette is

$$P_{\mu\nu}(x) = Q_\mu(x)Q_\nu(x + \hat{\mu})Q_\mu^\dagger(x + \hat{\nu})Q_\nu^\dagger(x). \quad (7.1)$$

With the implemented directed-overlap convention, $C_1^{\text{mesh}} = -(2\pi)^{-1} \sum_x \arg \det P_{12}(x)$. For four parameters write $F_{\mu\nu}(x) = \log P_{\mu\nu}(x)$ using the principal matrix logarithm. The non-Abelian estimator is

$$C_2^{\text{mesh}} = -\frac{1}{4\pi^2} \sum_x \Re \text{Tr}[F_{12}F_{34} - F_{13}F_{24} + F_{14}F_{23}](x). \quad (7.2)$$

The logarithms already include the discrete plaquette area. Mesh refinement, physical endpoint identifications, and a fixed isolated rank are essential. Coefficient arrays from different geometries cannot be contracted without cross-geometry AO overlaps. The numerical value is retained before rounding, and a sampled gap does not certify isolation between sampled geometries.

8 Spectral localizers in finite and periodic systems

8.1 Energy-position tuples and Clifford structure

A spectral localizer combines operators associated with several observables into one self-adjoint matrix. For two positions and a Hamiltonian, Pauli matrices provide a Clifford representation. In an orthonormal space define

$$\tilde{L} = (\tilde{H} - EI) \otimes \sigma_z + \kappa(\tilde{X} - xI) \otimes \sigma_x + \kappa(\tilde{Y} - yI) \otimes \sigma_y, \quad (8.1)$$

up to the explicitly fixed tensor ordering. The parameter $\kappa > 0$ has units of energy/length. It balances energy and position resolution rather than correcting the Hamiltonian. The connection between a localizer signature and an even index pairing holds under appropriate locality, gap, and finite-volume bounds [26]. Those hypotheses remain relevant when interpreting a finite Gaussian model.

In auxiliary-index-first ordering the covariant AO matrix is

$$L = \begin{pmatrix} H - ES & \kappa[(X - xS) - i(Y - yS)] \\ \kappa[(X - xS) + i(Y - yS)] & -(H - ES) \end{pmatrix}, \quad B = \begin{pmatrix} S & 0 \\ 0 & S \end{pmatrix}. \quad (8.2)$$

For SOC, H and S here denote the physical spinor matrices. Auxiliary doubling is separate from spin, so the localizer dimension is four times the scalar AO count. All AO directions are retained: there is no hidden occupied-space projection or automatic removal of nearly dependent basis functions.

8.2 Index, metric gap, and stability

For class A define

$$\nu_A = \frac{1}{2} \text{Sig}(L) = \frac{1}{2}(n_+ - n_-), \quad \mu = \min_j |\lambda_j|, \quad Lv_j = \lambda_j Bv_j. \quad (8.3)$$

Signature is invariant under nonsingular congruence, whereas ordinary eigenvalues of a covariant AO matrix are not. Inertia can therefore be evaluated directly from L , but the protection gap must retain B . Equivalently, μ is the smallest absolute eigenvalue of $B^{-1/2}LB^{-1/2}$; this identity does not require forming that dense matrix.

At fixed metric, the index cannot change under a perturbation satisfying

$$\|B^{-1/2}\delta LB^{-1/2}\| < \mu. \quad (8.4)$$

This finite-matrix statement gives local protection an operational meaning. If the metric changes, the corresponding orthonormalized problems must be compared, including the effect of δB . An unresolved near-zero gap yields an *unresolved* index, not evidence for trivial topology.

Moving the query and coordinate operators together, shifting the Hamiltonian and query energy together, or changing AO basis must leave the generalized gap unchanged. Reversing the oriented plane reverses the class A sign. Model comparisons establish the sign convention before applying it to a material.

8.3 Inertia and gap require different calculations

A pivoted Hermitian factorization has the form

$$P^\dagger L P = \mathcal{L} D \mathcal{L}^\dagger. \quad (8.5)$$

Here $\mathcal{L}$ is unit lower triangular and D contains 1×1 and 2×2 blocks. Sylvester's law transfers the inertia from L to D . For a Hermitian block,

$$D_b = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}, \quad \lambda_{\pm} = \frac{a+d}{2} \pm \sqrt{\left(\frac{a-d}{2}\right)^2 + |b|^2}.$$

Stable evaluation must retain scaling and both signs. A zero-diagonal block with $b \neq 0$ has one positive and one negative direction; counting its diagonal entries is wrong. The dense reference uses LAPACK ZHETRF, with DSYTRF for exactly real matrices [27]. A small imaginary part is not discarded to select real arithmetic.

Pivot magnitudes are not generalized gaps. For $\sigma > 0$, shifted inertia counts satisfy, away from singular endpoints,

$$\begin{aligned} n_-(L - \sigma B) - n_-(L) &= \#\{j : 0 < \lambda_j < \sigma\}, \\ n_-(L) - n_-(L + \sigma B) &= \#\{j : -\sigma < \lambda_j < 0\}. \end{aligned} \quad (8.6)$$

These counts bracket the nearest positive or negative generalized eigenvalue. Factorization failure or an ambiguous endpoint must not be treated as a successful certificate. The reported bounds are finite-precision numerical brackets, not interval-arithmetic guarantees.

8.4 Pfaffian orientation and physical time reversal

For a real skew-symmetric matrix A of even order, $\det A = \text{Pf}(A)^2$: the determinant loses the Pfaffian sign. Moreover, $\text{Pf}(P^T A P) = \det(P) \text{Pf}(A)$. An odd simultaneous row/column permutation reverses that sign without changing the determinant. Ordinary inertia cannot replace a skew factorization with explicit permutation bookkeeping.

For spinful time reversal with $\Theta^2 = -1$, the localizer has a real skew representation [28, 29]. In auxiliary/spin/AO ordering, let $C = J_{\text{aux}} \otimes J$ and $Q = (I - iC)/\sqrt{2}$. The symmetry relations imply that

$$A = iQ^\dagger L Q \quad (8.7)$$

is real and skew-symmetric. Both properties are checked numerically, with residuals small relative to the independently resolved metric gap. An arbitrary complex AO basis change also transforms time reversal; keeping J fixed would generally be inconsistent.

Let A_0 be the corresponding transform of $L_0 = \text{diag}(S, -S)$. The reference-calibrated index is

$$(-1)^{\nu_{\text{AII}}} = \text{sgn Pf}(A) \text{sgn Pf}(A_0). \quad (8.8)$$

Both factors use the same ordering and orientation. The dense algorithm accumulates signs rather than an overflowing Pfaffian product. This is a time-reversal local index, not an inversion-parity test.

The optional sparse skew-LDL backend implements a separate numerical task from MUMPS inertia [29]. Its present single-rank factorization must not be described as an MPI-distributed Pfaffian solver. The motivating photonic and aperiodic examples [9, 29] provide methodological context, not first-principles material validation for the CP2K path.

8.5 Finite three-dimensional class AII

For three Cartesian positions, the finite AII construction uses a real determinant rather than the two-dimensional Pfaffian [25]. In auxiliary-index-first ordering, define

$$\begin{aligned} A_3 &= I_2 \otimes (H - ES) - i\kappa \sum_{j=1}^3 \sigma_j \otimes (X_j - x_j S), \\ B_3 &= I_2 \otimes S, \quad L_3 = \begin{pmatrix} 0 & A_3 \\ A_3^\dagger & 0 \end{pmatrix}, \quad \mathcal{B}_3 = \text{diag}(B_3, B_3). \end{aligned} \quad (8.9)$$

Here X_j are the covariant Cartesian moments, x_j are the query coordinates, and the Pauli matrices σ_j act only in the auxiliary space. The physical-spin Hamiltonian, metric and all three positions must obey the fixed fermionic time-reversal representation. With $C = J_{\text{aux}} \otimes J$ and $Q = (I - iC)/\sqrt{2}$, the chiral block $R_3 = Q^\dagger A_3 Q$ is real but need not be symmetric. For a resolved gap, the local index is

$$\nu_3 = \frac{1 - \text{sgn det } R_3}{2}, \quad \mu_3 = \min_{\ell} |\lambda_{\ell}|, \quad L_3 v_{\ell} = \lambda_{\ell} \mathcal{B}_3 v_{\ell}. \quad (8.10)$$

Equivalently, μ_3 is the smallest singular value of $B_3^{-1/2} A_3 B_3^{-1/2}$. Nonsingular AO congruences multiply the determinant by a positive factor and preserve the metric gap. A basis change must also preserve or consistently transform the time-reversal representation.

The dense implementation accumulates triangular-pivot and permutation signs without forming a determinant product. The factor reconstruction residual and a rounding allowance are compared with the independently calculated metric gap. If excessive pivot growth prevents acceptance of the triangular factorization, a Householder factorization supplies orthogonal and upper triangular factors instead. The determinant sign includes every nonidentity reflection. The discarded imaginary part of R_3 is included in the perturbation budget, and an unresolved calculation returns no topological label.

The Hermitian pencil has order eight times the scalar AO count. Its current solver is a dense reference, including GPW/GAPW and post-SCF GTH SOC. The periodic extension is given below, while a sparse three-dimensional solver is not provided. This strong local index is not a collection of weak or two-dimensional slice indices. Agreement with a bulk invariant requires separate volume, basis and scale convergence. The Supporting Information includes an explicit finite-box counterexample to automatic identification with the bulk parity.

8.6 Periodic positions and the torus construction

A Cartesian coordinate truncated to a primitive cell is discontinuous at the periodic seam. Using it in the open-boundary localizer is not a periodic formulation. Instead, analytic periodic functions are used [30]. For analysis-torus reciprocal vectors $\mathbf{G}_j$ and $\phi_j = \mathbf{G}_j \cdot \mathbf{r}_0$, define

$$\begin{aligned} C_j &= \langle \phi_{\mu} | \cos(\mathbf{G}_j \cdot \mathbf{r} - \phi_j) | \phi_{\nu} \rangle, \\ T_j &= \langle \phi_{\mu} | \sin(\mathbf{G}_j \cdot \mathbf{r} - \phi_j) | \phi_{\nu} \rangle. \end{aligned} \quad (8.11)$$

In the implemented orientation,

$$L_{\text{per}} = \begin{pmatrix} H - ES - \eta(2S - C_1 - C_2) & \eta(T_1 - iT_2) \\ \eta(T_1 + iT_2) & -H + ES + \eta(2S - C_1 - C_2) \end{pmatrix}. \quad (8.12)$$

The energy scale η is distinct from κ , and the metric remains $\text{diag}(S, S)$. An overall sign and auxiliary-space conjugation relative to the cited periodic construction make its orientation consistent with the finite localizer used here.

The trigonometric operators are integrated directly, not evaluated as matrix functions of finite projected Cartesian positions. Projection need not preserve either exact unitarity of exponentials or their commutativity. The theorem for a gapped finite-range lattice Hamiltonian therefore does not automatically become a theorem for an arbitrary Gaussian projection. The output is an AO-torus diagnostic requiring independent basis-, torus-, and scale-converged material checks.

Explicit analysis k-points describe the enlarged torus through its complete mesh. Periodic exponentials shift between mesh sectors, so their cross-k links must be retained. A symmetry-reduced SCF can supply the potential, but scalar SCF weights cannot replace the full torus operator. For three periodic directions, the corresponding chiral block is

$$A_{3,\text{per}} = I_2 \otimes \left[H - ES - \eta \left(3S - \sum_{j=1}^3 C_j \right) \right] - i\eta \sum_{j=1}^3 \sigma_j \otimes T_j. \quad (8.13)$$

The phase-shifted C_j, T_j are the analytic AO integrals defined above, and $\eta > 0$ has energy units. Replacing A_3 by $A_{3,\text{per}}$ in Eqs. (8.9) and (8.10) gives the Hermitian metric pencil, real determinant sign and gap. In this orientation the chiral block is an energy-scaled adjoint of the odd-dimensional periodic construction of Ref. [30]. Taking the adjoint preserves the sign of its real determinant. The native three-dimensional path uses all three torus directions with fully periodic cell and Poisson boundaries, fermionic time reversal and the dense AII solver. Its finite-AO validation includes skew cells, origin periodicity and GPW/GAPW comparisons. It is not a sparse implementation or a bulk-convergence certificate.

Spectral flattening provides a separate way to examine the large bandwidth-to-gap ratio of a DFT Hamiltonian. For positive S and E in a resolved electronic gap, define

$$F_E = \Delta S \, \text{sign} \left[S^{-1}(H - ES) \right] = \Delta S^{1/2} \, \text{sign} \left[S^{-1/2}(H - ES)S^{-1/2} \right] S^{1/2}, \quad \Delta > 0. \quad (8.14)$$

This covariant Hermitian operator retains the occupied projector and replaces generalized band energies by $\pm\Delta$. The dispersion-reduction argument for periodic localizers motivates this control [30]. It does not equate a flattened finite-volume localizer with the original one: the localizer gap can close during flattening even when the electronic gap stays open. An exact sign is also generally longer ranged than H . Consequently, the material tests in the Supporting Information are distinguished from a proof of the finite-range theorem's hypotheses and from the native unflattened calculation.

The optional `SPECTRAL_FLATTENING` path evaluates Eq. (8.14) for finite systems with distributed DBCSR metric inversion and matrix-sign iterations. Before those iterations, the original positive metric and the distance of E to the generalized spectrum of (H, S) are checked by dense diagonalization or shifted MUMPS factorizations. An unresolved electronic gap is rejected rather than enlarged by flattening. This precondition is distinct from the gap of the final localizer. The transformed Hamiltonian is reused across position and scale queries at fixed E ; the SCF Hamiltonian and Gaussian coordinate operators remain unchanged. The option is off by default. The distributed inverse and exact-sign approximation can increase fill-in, so this path does not imply linear scaling.

For a translation-invariant periodic AO torus, the same sign can instead be evaluated in complete primitive Bloch blocks. The implementation verifies translation covariance of the entire Hamiltonian and metric before extracting a representative block row. It distributes full generalized

Bloch eigensystems over MPI ranks, forms $F_E(k) = \Delta S(k)C(k)\text{sign}(\epsilon_k - E)C(k)^\dagger S(k)$, and Fourier-transforms back to the original distributed spin/cell/atom order. The metric bounds and electronic gap refer to the union of all sampled Bloch spectra. No bands are discarded and no independent localizer is assigned to a single k-point. This avoids a whole-torus eigensystem or sign iteration for the flattening step; it does not remove the final localizer solve or its possible fill-in.

8.7 Connection to the quadratic pseudospectrum

For Hermitian operators A_j and anticommuting Hermitian Clifford matrices γ_j , squaring $L = \sum_j A_j \otimes \gamma_j$ gives

$$L^2 = \left(\sum_j A_j^2 \right) \otimes I + \sum_{j < l} [A_j, A_l] \otimes \gamma_j \gamma_l. \quad (8.15)$$

The first term measures simultaneous approximate localization; the second retains noncommutativity. The quadratic gap and localizer gap are related, but not interchangeable. In an AO basis each projected product must include the metric, or be evaluated after orthonormalization.

9 Quadratic pseudospectrum and localized approximate states

The quadratic pseudospectrum asks for a normalized state that minimizes the combined residual of incompatible observables [31]. For a query $(E, \mathbf{x})$, set $A_0 = H - ES$, $A_j = X_j - x_j S$, $w_0 = 1$, and $w_j = \kappa$ for $j = 1, 2, 3$. The AO problem is

$$Qc = \lambda Sc, \quad Q = \sum_{j=0}^3 w_j^2 A_j^\dagger S^{-1} A_j, \quad c^\dagger Sc = 1. \quad (9.1)$$

The minimum $\sqrt{\lambda}$ is the quadratic gap. Its eigenvector is a joint approximate state, and several low eigenpairs can describe a localized subspace. The separated residuals are

$$\begin{aligned} \delta E^2 &= (A_0 c)^\dagger S^{-1} (A_0 c), \\ \delta x_j^2 &= (A_j c)^\dagger S^{-1} (A_j c), \\ \lambda &= \delta E^2 + \kappa^2 \sum_j \delta x_j^2. \end{aligned} \quad (9.2)$$

Expectations $c^\dagger H c$ and $c^\dagger X_j c$ distinguish bias from spread around the query. These states need not be eigenstates of H . Within degenerate minimizing spaces, projectors or subspace observables must be compared instead of individual coefficients.

Equation (9.1) is positive semidefinite and covariant under Eq. (2.3). Ordinary matrix products without metric solutions would destroy that invariance. The position residual concerns the projected position: $X_j S^{-1} X_j$ is not an independently integrated continuum $\langle r_j^2 \rangle$. Small bases can underestimate continuum localization errors even when the eigenproblem is solved accurately.

The dense reference uses a complete orthonormalized space. The iterative path applies Q without forming it: multiply by A_j , solve with S , multiply by $A_j^\dagger$, and accumulate. This avoids fill-in from explicitly squaring sparse matrices. A complex block Rayleigh–Ritz iteration, double metric reorthogonalization, and thick restarts address clustered low states. The residual

$$\|Qc - \lambda Sc\|_{S^{-1}} \leq \epsilon_{\text{eig}} \max(1, |\lambda|)$$

checks an eigenpair in atomic units; it does not independently prove that no lower state was missed.

Reusable MUMPS factors avoid forming S^{-1} , but the present trial vectors and multiple-right-hand-side communication are not fully distributed. Output includes separated residuals, expectations, and optional complex AO coefficients. The finite analysis covers isolated GPW/GAPW systems and post-SCF GTH SOC. A small quadratic gap is a localization diagnostic, not a Chern or $\mathbb{Z}_2$ index.

9.1 Periodic joint localization on an AO torus

The periodic formulation replaces the unbounded position by paired trigonometric coordinates on a Born–von Karman torus. For cell matrix h and a full Gamma-centered $N_1 \times N_2 \times N_3$ property mesh, define, for each periodic lattice direction j ,

$$g_j = \frac{2\pi}{N_j}(h^{-1})_{j,:}, \quad r_j = |g_j|^{-1}, \quad (9.3)$$

$$C_j = \langle \cos(g_j \cdot r) \rangle_{\text{AO}}, \quad D_j = \langle \sin(g_j \cdot r) \rangle_{\text{AO}}.$$

These covariant matrices are assembled from the same ordered Berry integrals and real-cell images as the periodic localizer. For query x , replace each Cartesian observable in Eq. (9.1) by the two shifted operators $C_j - \cos(g_j \cdot x)S$ and $D_j - \sin(g_j \cdot x)S$, both weighted by κr_j . With their dimensionless residuals δC_j and δD_j , the resulting problem satisfies

$$\lambda = \delta E^2 + \kappa^2 \sum_{j \text{ periodic}} \delta \ell_j^2, \quad \delta \ell_j = r_j \sqrt{\delta C_j^2 + \delta D_j^2}. \quad (9.4)$$

The length-valued chord residual is invariant under shifting the query by a torus lattice vector. For skew cells this is a specified reciprocal-coordinate metric, not an isotropic Cartesian variance; changing the primitive-lattice basis can change that metric. Finite AO projection also means that $C_j + iD_j$ need not be exactly unitary. We do not replace it by a polar-unitarized operator.

The **FORMULATION PERIODIC** path uses the complete torus AO space at the frozen SCF potential, independently of SCF symmetry weights or reconstructed occupied MOs. It therefore accepts a symmetry-reduced SCF calculation without claiming an irreducible property eigensolver. One, two or three periodic directions are supported; open directions are not localized by this formulation. Output reports chord residuals and mean cosines/sines rather than ambiguous Cartesian mean positions. Dense and DBCSR/MUMPS solvers and the restricted post-SCF GTH-SOC construction share this tuple of Hermitian operators. Translation-operator band unfolding is a different joint-eigenvector problem, discussed separately below.

Small GPW/GAPW He and GTH-SOC Ne checks give serial-dense versus two-rank-iterative gaps agreeing within 6×10^{-15} hartree; the maximum iterative eigenpair residual is 6.7×10^{-11} hartree². The He primitive-mesh and separately converged Gamma-supercell gaps differ by less than 10^{-9} hartree. Query wrapping, rigid cell rotation and equivalent SCF mesh schemes provide additional consistency checks. Settings and limitations are given in the Supporting Information; these are numerical integration tests, not converged material localization predictions.

9.2 Complex translation queries: numerical validation

Joint Hamiltonian/translation residuals provide a route to momentum-resolved approximate states [32]. The numerical kernel also accepts complex queries and nonnormal covariant operators. For a translation matrix T and $z = \exp(i\mathbf{k} \cdot \mathbf{a})$, the generalized eigenproblem $Q_{E,z}c = \lambda Sc$ uses

$$Q_{E,z} = (H - ES)S^{-1}(H - ES) + \kappa_T^2(T - zS)^\dagger S^{-1}(T - zS), \quad (9.5)$$

where κ_T has units of energy. DBCSR applies the forward and adjoint operators; reusable metric factors supply the solves. The adjoint is essential: for $T = C + iD$ in an orthonormal basis, the translation normal product contains $i[C, D]$. Independent Hermitian squares of C and D omit this term for nonnormal T .

An 18-site trimerized chain reproduces all 1,170 eigenvalues from independently assembled Bloch blocks over 65 queries within $1.6 \times 10^{-14}t^2$, with t the model energy scale. An open-chain counterexample distinguishes the correct normal product from the Hermitian-square approximation by $8.60 \times 10^{-2}t^2$. Complex basis-change, restarted-iteration and distributed adjoint tests are detailed in the Supporting Information. These are model-kernel checks, not DFT unfolding results or parallel scaling measurements.

9.3 Finite Gaussian Hamiltonian/translation analysis

For isolated GPW/GAPW systems, the Gaussian adapter evaluates $T_{\mathbf{a}} = \Phi^\dagger U(\mathbf{a}) \Phi$ analytically, using $U(\mathbf{a})\psi(\mathbf{r}) = \psi(\mathbf{r} + \mathbf{a})$ and shifted ket centers $\mathbf{R}_B - \mathbf{a}$. Pairs are screened at their translated separation, not restricted to the unshifted SCF overlap list. With $S = \Phi^\dagger \Phi$ and $z_{\mathbf{a}} = e^{i\mathbf{k} \cdot \mathbf{a}}$, the exact continuum translation normal Gram is

$$N_{\mathbf{a},z} = (1 + |z|^2)S - \bar{z}T_{\mathbf{a}} - zT_{\mathbf{a}}^\dagger. \quad (9.6)$$

Replacing the translation term in Eq. (9.5) by $\kappa_T^2 \sum_{\mathbf{a}} N_{\mathbf{a},z_{\mathbf{a}}}$ retains the part of the translated wavefunction outside the AO span. For each normalized state, the full and projected residuals obey

$$\delta U_{\mathbf{a}}^2 = \delta U_{\mathbf{a},\text{AO}}^2 + \ell_{\mathbf{a}}^2, \quad \ell_{\mathbf{a}}^2 = c^\dagger (S - T_{\mathbf{a}}^\dagger S^{-1} T_{\mathbf{a}}) c \geq 0. \quad (9.7)$$

The output reports all three terms; no polar unitarization is used. The energy residual still concerns the projected AO Hamiltonian, not an independently integrated continuum H^2 .

Independent shifted ghost-basis calculations verify the spectra, complex expectations and leakage decomposition for He, H₂ and Ne/GAPW, with up to 17 AOs and TZVP MOLOPT functions. A normalized single-Gaussian reference additionally has an analytic translation overlap. The largest quadratic-eigenvalue discrepancy is 6.1×10^{-12} hartree² across serial dense and two-rank iterative tests. Translation leakage is substantial in these deliberately small spaces, reaching 0.87 for the single Gaussian; it cannot be discarded on the basis of solver convergence. The SI supplies settings, decomposition checks, and additional four-rank all-electron and GTH-SOC comparisons.

The FORMULATION TRANSLATION input accepts Cartesian translation vectors and wave-vector queries. The isolated-system path imposes no boundary weighting; finite-size and basis effects must therefore be assessed separately.

9.4 Periodic Gaussian translation checks

The same continuum normal Gram can be evaluated in a periodized Gaussian basis on a Born–von–Karman torus. The analysis mesh specifies all real-cell copies; image-resolved Hamiltonians and overlaps are lifted at the converged SCF potential. Translations are evaluated by summing displaced Gaussian overlaps over torus images, with reciprocal-plane bounds that also cover skew cells. SCF symmetry reduction does not reduce this property space. For an exact lattice translation represented by a cyclic AO permutation P , $T = SP$ and the leakage in Eq. (9.7) vanishes. Arbitrary translations generally leave the finite AO span and retain nonzero leakage.

Independent periodic ghost-basis calculations test He/TZVP on cubic and skew cells, using GPW and GAPW. The lowest three quadratic eigenvalues agree with a separately assembled normal Gram within 8×10^{-16} hartree²; independently converged primitive $3 \times 1 \times 1$ meshes and explicit three-cell Gamma supercells agree in their quadratic gaps within 10^{-10} hartree. This comparison requires preserving the converged hard and soft onsite GAPW densities when rebuilding the frozen image-resolved potential. The complete 18-state lattice-translation spectrum and the sign of its complex phases are also checked. The SI gives these references, rotation and wrapping controls, and dense/iterative one- and two-dimensionally periodic tests, including post-SCF GTH SOC.

A diamond-Si control extends this comparison to a bonded crystal with 102 TZVP AOs on a three-cell torus. For a primitive lattice translation, the quadratic spectrum can be predicted directly from the ordinary CP2K Bloch band energies and translation phases. All 105 tested low eigenvalues agree with that reference within 1.3×10^{-10} hartree², limited in part by the printed band precision. Serial dense and two-rank iterative gaps agree within 7×10^{-12} hartree. An independently converged six-atom Gamma supercell agrees within 6×10^{-12} hartree. These comparisons expose an important numerical requirement: SCF convergence alone does not certify Hermiticity of the image-resolved AO Hamiltonian. Tightening nonlocal-projector screening

reduces the measured relative Hermiticity defect from 4.1×10^{-8} to 5×10^{-14} , without modifying the torus transformation or relaxing its checks. The SI retains the failed controls as well as the successful band comparison.

Displacing one Si atom by 0.1 Å breaks the primitive translation in the same six-atom cell. Separate shifted-ghost overlap integrals then replace the exact permutation reference. The 105 low eigenvalues agree with the independently assembled quadratic operator within 3.8×10^{-14} hartree². The selected states now have nonzero translation leakage, 0.00365–0.06825, and gaps change by up to 0.0133 hartree. This checks the noncommuting, incompletely represented translation case rather than assigning Bloch labels to a distorted crystal.

The frozen potential must also retain every density field used by the chosen functional. A separate He₂/TZVP control compares all twelve post-SCF Gamma band energies with an independent generalized diagonalization of the original SCF Hamiltonian and overlap. Preserving the kinetic-energy density for meta-GGAs and the separate GAPW_XC density reduces discrepancies of up to 1.56×10^{-2} hartree to below 1.9×10^{-10} hartree, the precision of the printed band energies. The SI distinguishes this shared frozen-potential check from validation of each property adapter.

These calculations validate the periodic translation adapter, not a converged material-unfolding spectrum. The quadratic gap is a joint energy/translation diagnostic, not a band spectral weight or a topological invariant. Physical translation choices, basis completeness, momentum resolution and defective-supercell size convergence remain separate scientific questions.

10 Materials-discovery applications and demonstrative tests

10.1 Hybrid-functional topological materials databases

TopoHSE-DB provides an application example of electronic-structure-based topological classification at database scale [33]. Mirhosseini, Elcoro, Knüpfer and Kühne combine consistently relaxed nonmagnetic crystal structures, PBE and screened-hybrid HSE06 electronic structures, and topological quantum chemistry. Comparing the resulting labels exposes the sensitivity of a symmetry-based classification to the underlying exchange–correlation approximation. A converged integer or symmetry label is therefore not, by itself, evidence that the electronic Hamiltonian has been described accurately.

The original database workflow uses VASP, VASP2Trace, and CheckTopologicalMat. It is cited here as a materials-discovery application and a source of future comparison cases, not as a calculation performed with the new CP2K topology routines. Connecting such a workflow to CP2K requires equivalent electronic structure and symmetry conventions as well as the general little-group, compatibility-relation, and EBR machinery discussed in Section 6. The currently implemented inversion analysis does not yet reproduce the database’s full classification workflow.

10.2 Generative–predictive discovery of candidate materials

The submitted study by Brocai, Kühne, Knüpfer and Mirhosseini, *Generative–predictive models enable the discovery of topological materials*, provides a complementary application [34]. It couples topology-conditioned crystal generation to a predictive filter using a shared pretrained representation, followed by hybrid-functional electronic-structure and TQC analysis. The submitted manuscript reports 1,033 successfully calculated hybrid-functional band structures; 629 receive an unambiguous classification and 523 are classified as topological. The corresponding 83.15% rate applies to the classified subset, whereas the confirmed yield across all 1,033 calculated cases is 50.6%. The 404 unresolved classifications must not be silently counted as either trivial or topological.

These results, obtained with the study’s VASP-based workflow, illustrate the demand for reliable post-processing and explicit failure diagnostics. They also motivate subsequent, distinct questions: Wilson spectra can probe isolated band subspaces, Kubo calculations can address

transport at specified temperature and dissipation, and localizers can investigate finite, defective, or disordered realizations of selected candidates. These are prospective CP2K follow-up applications, not calculations already completed for the reported candidate set. A TQC label alone does not supply a conductivity or a localizer protection gap.

10.3 Small calculations as tests and demonstrations

The smaller calculations in this paper and the companion k-point study serve a different purpose [35]. They provide transparent tests and demonstrations of individual algorithmic steps, not a second materials-discovery survey. Table 1 states the role of each set.

Calculation set	Demonstration and test objective
Si and Al, companion k-point study	Quadrature dependence and equivalent full-grid, time-reversal, K290, and SPGLIB representations. At fixed 4^3 sampling, 64 eigensystems reduce to four with atomic symmetry.
Periodic neon and buckled stanene	Trivial and nontrivial Wilson-loop examples, with $\nu = 0$ and $\nu = 1$, respectively; comparison of native and external analyses.
Ne/SOC, Bi ₂ /SOC, and diamond/SOC	Primitive-mesh versus supercell and reduced versus full-property-grid checks of the metric, Hamiltonian, and current operators.
Analytic localizer models and small GPW/GAPW systems	Known indices, gap closings, basis covariance, and dense/sparse solver consistency; quadratic-state residual and subspace checks.

Table 1: Test and demonstration calculations, distinct from the materials-discovery applications cited above. The Si/Al results are retained from the companion study, not rerun or reassigned to the new topology implementation.

For Si and Al, the largest fixed-mesh cell-energy or electronic-free-energy difference among the compared representations is 1.4×10^{-12} hartree. This tests representation equivalence, not complete basis or k-point convergence. In particular, the Al free energy still changes by 10.721 meV per atom between the 6^3 and 8^3 meshes. Likewise, agreement between Wilson analyses using the same overlap matrices tests the analysis stage, while independent operator identities test the AO construction. The quantitative checks in Section 11 and the Supporting Information retain these distinctions and their original calculation provenance.

11 Implementation, validation, and development boundaries

11.1 Shared operators and optional numerical backends

The finite-system property paths share full AO Hamiltonians, overlap, analytic positions, and optional GTH-SOC blocks. DBCSR supplies distributed block storage and matrix products [36]. Localizer assembly retains the union of the operator sparsity patterns without additional dropping. DBCSR is not itself a general pivoted indefinite factorization or Pfaffian solver.

The dense LAPACK path is the small-system reference. MUMPS supplies distributed symmetric-indefinite factorizations and reusable metric solutions [37]. Complex Hermitian matrices are realified:

$$\mathcal{R}(A) = \begin{pmatrix} \Re A & -\Im A \\ \Im A & \Re A \end{pmatrix}. \quad (11.1)$$

Each eigenvalue is duplicated, so inertia-based formulas compensate for that duplication. A positive metric is checked separately from the indefinite localizer. Static pivot perturbations and low-rank compression are not silently accepted as exact sign counts. Symbolic analysis is reused across appropriate shift families.

The optional Tacho skew-LDL extension is accessed through a small C interface and Fortran interoperability [42]. No MATLAB dependency is introduced. Permutation parity, skew-pivot conventions, numerical residuals, and the trivial-reference orientation are explicit. Its sparse factorization currently resides on one rank, while DBCSR assembly and MUMPS metric/gap checks can be distributed. PARDISO and PEXSI are possible alternative backends, not prerequisites and not part of the tested backend set reported here.

MUMPS factor storage and process-summed solver memory are not the whole application’s memory consumption. Sparse fill-in, source-rank coordinates, dense reference arrays, trial vectors, and MPI buffers must be accounted for separately. Similarly, single-node timings cannot establish multi-node scaling.

11.2 Three levels of validation

The first level tests algebra independently of an SCF calculation: metric congruences, origins, energy shifts, degeneracies, time reversal, permutation orientation, and analytic model indices. The second compares numerical algorithms on identical matrices: dense eigenvalues against LDL inertia, dense against sparse gap brackets, dense against skew-LDL signs, and full against iterative quadratic eigenproblems. The third exercises operator construction from actual converged CP2K GPW/GAPW calculations.

Retained checks include open and periodic Chern models, spin-mixed time-reversed model pairs, gap closings, singular and indefinite metrics, and deliberately broken symmetry. They establish nontrivial topology in controlled models. Small molecular and bulk integration fixtures instead establish that the AO data reach the algorithms consistently; most are not nontrivial first-principles topological materials.

The retained neon and stanene Wilson results in Section 5 are small band-topology test and demonstration calculations, with matching inversion indices. They complement the Si/Al k-point demonstrations summarized in Section 10, rather than constituting a materials-discovery application study. They were produced with their recorded executable and sampling. They are not relabelled as localizer or Kubo calculations merely because the underlying operators are now shared.

Separate source-frozen checks completed on 1 October 2026 reproduce the original Wilson examples and rebuild the localizer and quadratic paths from recorded sources. After two targeted corrections, the latter pass 12 unit-test executions and all 124 regression checks in each of two MPI/OpenMP configurations. The corrections retain numerical references and bounds checking. They address premature iterative-subspace restarts and a compiler-dependent GAPW array-section failure. A separately rebuilt native Pfaffian adapter reproduces 33 archived material-matrix queries without rerunning their SCF calculations. The Supporting Information records the source boundaries, initial failures, independent controls, and dependency fingerprints. Neither the single-node runs nor the matrix replays establish multi-node scaling or additional material convergence.

11.3 Quantitative test and demonstration calculations

Table 2 records representative comparisons from the retained development runs. Agreement is measured between calculations at identical numerical settings unless noted otherwise. These are consistency checks, not a claim of basis- and material-converged conductivity.

For the diamond property reconstruction, metric, Hamiltonian, and current residuals are checked separately rather than inferred from the final scalar conductivity. Degenerate subspaces retain their transformed gauge. The representative count measures avoided diagonalizations, not total speedup.

The periodic localizer comparison uses the same Born–von Karman torus represented in two ways, not a weighted sum of independent irreducible-k localizers. Both localizers have order 312, with gaps 0.380994261391 and 0.380994261819 hartree. The Ne calculation is a trivial integration

Comparison	Retained result and meaning
Ne/SOC primitive mesh versus Gamma supercell	A $3 \times 1 \times 2$ primitive mesh and its explicit Gamma supercell give $6.21105320 \times 10^{-9}$ S at the converged comparison settings.
Bi ₂ /SOC property symmetry	Six of 27 eigensystems suffice; the maximum printed full-tensor relative difference from the unreduced calculation is 2.71×10^{-15} .
Primitive diamond/SOC	Both K290 and SPGLIB use three of eight representatives; the full-tensor relative difference is 2.21×10^{-11} .
Periodic Ne/SOC localizer	A $3 \times 1 \times 2$ primitive mesh and the corresponding Gamma supercell give $\nu = 0$; their generalized gaps differ by 4.28×10^{-10} hartree.
Spin-mixed model localizers	Dense and sparse Pfaffian paths distinguish $\nu = 1$ and $\nu = 0$; the nontrivial generalized gap is unchanged by a complex nonorthogonal basis transformation at printed precision.
Bi ₂ quadratic states	Dense and iterative gaps agree within 6×10^{-15} hartree; four metric-subspace singular values differ from unity by less than 10^{-10} .

Table 2: Operator-level checks retained on 24 September 2026. Raw settings and limitations are recorded in the Supporting Information. The transport numbers are numerical fixtures, not material predictions.

fixture, not a nontrivial material classification. For the spin-mixed model, the nontrivial and trivial gaps are 0.79004778 and 1.1109451 in model energy units; sparse skew-factor solve-probe residuals are below 2.3×10^{-14} . These probes complement the independently computed gap and Pfaffian orientation; they are not by themselves a global factorization-error bound.

A separate two-dimensional stanene calculation tests operator consistency as well as the distinction between a resolved finite matrix and a bulk-material classification. Post-SCF GTH-SOC Wilson surfaces give $\nu = 1$, with sampled indirect gaps of 0.07369 and 0.07657 eV for DZVP/200 Ry and TZVP/400 Ry, respectively. Reconstructing the full 3×3 -torus localizer from all Bloch spinors and complex neighbour overlaps reproduces its direct AO construction to 1.85×10^{-12} hartree. This cross-path test identified and removed a redundant SOC atom-block sign in the torus assembly; agreement between dense and sparse solvers alone did not detect it. The corrected $\eta = 0.03$ hartree query has $\nu = 0$ and gap 6.26487×10^{-4} hartree, while the 4×4 query has $\nu = 1$. Thus these tests do not establish a size-converged stanene localizer classification. The Supporting Information reports the corrected queries, the full-matrix comparison and the provenance of the superseded development data separately.

A subsequent full-band reference applies Eq. (8.14) without altering the Gaussian coordinate links or discarding empty states. With flattened energies ± 1 hartree and $\eta/\Delta = 0.75$ or 1, the 6×6 , 8×8 and 9×9 DZVP tori all give $\nu = 1$; the 6×6 TZVP control agrees. The $\eta/\Delta = 1$ gaps are 0.062925, 0.024216 and 0.186946 hartree for the three DZVP sizes, and 0.064210 hartree for TZVP. These are gaps of the rescaled localizer, not electronic band gaps. The small-torus and $\eta/\Delta = 1.25$ counterexamples remain in the Supporting Information. Thus flattening improves the tested index consistency, but the nonmonotonic gaps and residual scale dependence do not establish asymptotic convergence. The independent full-band eigensystem analysis also provides a reference for the native DBCSR matrix-sign implementation. The native DZVP 3×3 and 6×6 comparisons, and a TZVP 6×6 control, reproduce the selected gaps within 6×10^{-13} hartree and reproduce every tested index, including the small-torus counterexamples. A separate MUMPS/Tacho comparison and finite GPW/GAPW/all-electron checks are detailed in the Supporting Information. This verifies the implementation, not the convergence of the material classification.

A complementary sparse complete-band representation extends the flattened stanene control to 12×12 , 15×15 and 18×18 tori, without replacing the coordinate links by unitaries. At

the sampled scales $\eta/\Delta = 1, 1.25, 1.5$ all three sizes give $\nu = 1$. The corresponding $\eta/\Delta = 1$ gaps are 0.229914, 0.252124 and 0.268152 hartree. At $N = 12$, TZVP/400 Ry changes this gap to 0.231852 hartree; refining the scalar SCF mesh from 8×8 to 12×12 with DZVP/200 Ry gives 0.229840 hartree. Both controls retain $\nu = 1$. The representation reproduces 15 previously dense reference gaps within 4.3×10^{-15} hartree, with every resolved sparse parity agreeing. Some other scale queries fail the sparse-factor checks despite a resolved spectral gap; these are reported as unresolved rather than assigned the expected integer. A separately retained correction to floating-point pivot comparisons resolves a small-torus failure and is covered by an exact-Pfaffian regression. It does not supply delayed pivoting between elimination blocks or establish general solver robustness. An additional safety check uses independent solve right-hand sides rather than vectors of the form $b = Ax$, which can hide nullspace errors. Analytically rank-deficient controls and their scale/permutation variants verify the resulting rejection; this does not replace the separate spectral-gap calculation. A subsequent serial research extension passes unstable skew pivots and their Schur contributions to parent fronts without restarting symbolic analysis. It resolves all 33 retained stanene queries, including the five unresolved cases in the floating-pivot-corrected scan, with maximum independent-solve residual 2.06×10^{-11} at the unchanged 10^{-10} threshold. All 15 available dense references agree. Separate synthetic tests check Pfaffian signs, singular rejection, permutations and reconstructed factors. Subsequent native integration adds an explicit numerical-buffer limit and collective error propagation. A Bismuth SOC integration failure exposed one-sided roundoff in the skew matrix’s sparsity pattern; a consistent real-skew projection, bounded relative to the independently determined gap, removes this failure without changing reference values. The finite and periodic localizer directories pass 74 checks with four MPI ranks, and 45 checks without the optional sparse libraries. Source snapshots, prototype and native results, and memory limitations are retained separately in the Supporting Information. Numerical Pfaffian factors are still stored on a single MPI rank. Fresh one-parameter controls separate discretization effects at fixed 12×12 analysis size. With DZVP, cutoffs of 200, 400 and 600 Ry change the $\eta/\Delta = 1$ localizer gap by less than 4.4×10^{-7} hartree. At the common 400 Ry cutoff, switching from DZVP to TZVP instead changes this gap by 1.94×10^{-3} hartree. With TZVP/600 Ry, refining the scalar SCF mesh from 12×12 to 18×18 changes it by 6.9×10^{-6} hartree. All three sampled localizer scales retain $\nu = 1$. These controls distinguish a stable sampled index from quantitative basis and volume convergence; the Supporting Information retains the complete exports and separate electronic gaps. The independent DZVP 21×21 torus also gives $\nu = 1$ at all three sampled scales. Its $\eta/\Delta = 1$ gap is 0.282052 hartree, still 5.2% above the 18×18 value. A fresh sampling-converged TZVP/600 Ry Wilson surface independently confirms the nontrivial band invariant. Thus the expanded evidence supports consistency over more sampled sizes and scales, but neither quantitative gap convergence nor the finite-range theorem’s full hypotheses. The Supporting Information distinguishes this reference representation from the native real-cell solver.

A second SOC material reference uses a fixed Bi(111) bilayer geometry with $a = 4.33$ Å and buckling 1.74 Å from Ref. [41]. The restricted PBE potential and complete 26-state scalar DZVP space define a post-SCF GTH-SOC Hamiltonian. Its occupied ten-spinor Wilson surface converges to $\nu = 1$ after three refinements, with a final sampled indirect electronic gap of 0.403202 eV. The scalar SCF mesh is $8 \times 8 \times 1$, whereas the final Wilson surface contains 97 loops with 96 points per loop. This provides an independent periodic reference for finite-flake tests, not a size-converged finite localizer result or a reproduction of the reference article’s different electronic Hamiltonian.

A separate finite-system check uses an unrelaxed 18-atom patch of this geometry, with 234 scalar AOs and all 468 spinor states. Across 27 position–energy–scale queries, the native sparse localizer and an independently reconstructed dense reference agree on every $\mathbb{Z}_2$ index. The largest distance of the independent gap from the native inertia bracket is 2.8×10^{-12} hartree. Direct comparison of the three AO SOC components exposed and corrected a one-image MPI storage error in the post-SCF export: desymmetrization had moved offsite blocks away from the ranks owning their neighbour-list entries. After the correction, the finite and exported components agree

within 8×10^{-16} hartree. These checks validate the operator mapping and numerical analysis; the bare, thermally smeared flake is not an edge-relaxed, size-converged bulk-topology prediction.

A subsequent neutral-gap scan compares the same patch at scalar SCF smearing temperatures of 300 and 30 K. The electronic gap decreases from 7.74 to 2.18 meV, while both frontier Kramers doublets remain predominantly boundary weighted. At each neutral gap midpoint and the patch centroid, both Hamiltonians have $\nu = 1$ throughout $0.0025 \leq \kappa \leq 0.0035$ hartree/bohr, using an adaptive numerical gap bound rather than interpolation of sampled indices. All 45 native 30 K queries also agree with the independent dense reference. These results establish a stable finite-model scale window, not convergence to an insulating thermodynamic limit; the construction, precision margins and memory measurements are specified in the Supporting Information.

A controlled termination change removes only the two singly coordinated corner atoms, retaining the cell and all other coordinates. At 300 K the resulting neutral 16-atom patch has a 219.6 meV electronic gap and retains a nontrivial central localizer on the common subinterval $[0.00275, 0.0035]$ hartree/bohr. Its broader scale interval contains a localizer gap closing and is explicitly rejected. This comparison demonstrates sensitivity to termination, not edge relaxation or convergence with system size.

Increasing the same 16-atom calculation from UZH DZVP to TZVP changes its gap from 219.6 to 231.7 meV. Physical cross-basis overlaps, rather than direct AO-coefficient differences, give a largest occupied-subspace principal-angle sine of 0.1048; the mean retained occupied weight is 99.687%. One of the 45 sampled localizer indices changes, and the central scale-window boundary moves. Both bases retain $\nu = 1$ on the smaller common interval $[0.003, 0.0035]$ hartree/bohr at their own neutral midpoints. Thus gauge-independent subspace comparison and localizer stability provide complementary controls, without implying complete-basis convergence of this bare finite model.

The separate 30-atom DZVP control reinforces this distinction. Its neutral gap is 72.14 meV and its frontier states remain strongly boundary weighted. At the centroid and its own neutral midpoint, the entire interval $[0.003, 0.0035]$ has $\nu = 0$, whereas the 16-atom calculations have $\nu = 1$. The larger patch instead has resolved nontrivial intervals at smaller scales. A finer scan also finds a second nontrivial region missed by the original coarse grid. Consequently, neither one chosen scale nor agreement on a finite set of queries establishes a size-converged material classification.

Extending the same bare-flake series to 48 atoms reduces the neutral gap further to 37.92 meV. Boundary atoms constitute 37.5% of this patch but carry 65.14% and 64.49% of the occupied and empty frontier-doublet weights. This finite-flake level gap is not the periodic bulk gap; moving localizer-window boundaries alone do not imply a bulk phase change. At three selected scales, the native sparse calculation and a separately integrated complete-AO reference agree on every index, with each reference gap inside its native MUMPS bracket. This is a three-query native check, distinct from the independent 45-query survey.

A separate electronic-smearing control holds this 48-atom model fixed and lowers the scalar-reference temperature from 300 to 100 K. Its neutral gap changes from 37.92 to 36.81 meV, while all indices in the 45-query survey and the 20-scale centroid scan are unchanged. The occupied-subspace projector distance is nevertheless 0.05181, so unchanged sampled indices do not mean an unchanged SCF solution. Fresh gap bounds also retain the three resolved 48-atom parameter boxes discussed below, separately at 100 K. These two temperatures alone do not establish the zero-temperature limit; the physical-metric comparison is detailed in the Supporting Information.

The concentric 96-atom control extends the quantitative validation to a larger finite system and measures its fixed-geometry basis dependence. Its neutral SOC gap is 17.87 meV with UZH DZVP and 21.87 meV with UZH TZVP. The occupied 480-spinor subspaces have projector distance 0.1511 despite a mean retained weight of 99.70%. At the centroid and each calculation's own neutral midpoint, eight of nine sampled localizer indices agree between the bases. At $\kappa = 0.002$ hartree/bohr the index instead changes from zero to one. Four native TZVP controls agree with

the independent complete-AO reconstruction within 3.02×10^{-11} hartree, including this changed index. The TZVP result is thus independently validated at the chosen basis and geometry, and the comparison quantifies the sensitivity of the finite-flake classification to the orbital basis.

This contrast persists in joint parameter neighborhoods, not only along a line in κ . A norm bound for simultaneous changes in energy, both in-plane positions and scale keeps the finite localizer gapped for energies within 2% of each calculation’s neutral gap around its midpoint and positions within 0.05 Å of its centroid in each in-plane direction. On [0.003, 0.0035] hartree/bohr, both 16-atom bases retain $\nu = 1$, whereas the 30- and 48-atom DZVP models retain $\nu = 0$ throughout their respective boxes. Two lower-scale boxes for the 30-atom flake retain $\nu = 1$. They do not transfer in full to 48 atoms, where both contain opposite-index samples; freshly checked, narrower subboxes do retain $\nu = 1$. Adaptive coverage, independent full-spectrum corner checks and the numerical margins are detailed in the Supporting Information. The Hamiltonian is frozen in each box; these tests do not change the SCF potential or establish size convergence.

The retained open-boundary Qi–Wu–Zhang model measurements also illustrate when sparse factorization becomes useful. For a 30×30 lattice, the complex localizer has order 3,600 (7,200 after realification). A source-rank LAPACK reference requires 23.24 s, compared with 1.37 s for four-rank MUMPS including metric validation and shifted gap factorizations. In contrast, for 10×10 the dense calculation is faster (0.034 versus 0.057 s). For the larger case, increasing MUMPS from one to four ranks reduces time from 2.40 to 1.37 s but raises its process-summed solver memory from 19 to 45 MB. These are single-run, single-node kernel timings, not a comparison with distributed dense diagonalization, whole-process memory measurements, or a demonstrated multi-node scaling law. The full timing matrix, near-gap rejection tests and molecular quadratic-state diagnostics are retained in the Supporting Information.

The crystalline representation tests cover all 530 Hall settings of the 230 ordinary space groups, scalar and spinful factors, and their grey extensions. They validate 6,360 unitary projective character tables, 5,324 antiunitary corepresentation tables, and 12,720 directed segment comparisons; 4,462 segments retain antiunitary symmetry. Release and bounds-checked debug runs agree. The Gaussian little-group directory passes 69 numerical assertions in both serial and two-rank MPI runs, including GPW, GAPW, post-SCF SOC, screw monodromy and refined Wilson links. These counts describe algebra and small integration tests, not 230 separate converged material studies. The Supporting Information separates this evidence from a supplied magnetic-group algebra sweep and its independent reference comparison.

The site-induction kernel adds 38,160 site/k-point cases over the same Hall settings, using three positions per cell. Their 67,824 induced local-representation columns have valid little-group decompositions and satisfy the checked directed compatibility relations. Independent Cartesian s/p/d orbital traces, with and without physical spin, give 114,480 matching character comparisons. This tests atomic reference construction rather than a Wyckoff catalogue or an EBR classification of first-principles bands.

The separate geometric enumeration generates 3,467 Wyckoff families over all 530 Hall settings. Every family matches a tabulated affine coordinate family bijectively; the largest matching residual is 2.22×10^{-16} . The independent comparison also verifies 5,648 directed specialization relations. Each generated site is accepted by the spinful induction kernel with the correct multiplicity and stabilizer. This supplies geometric reference positions, not an elementary-band classification of electronic-structure results.

Combining all generated sites into matrices at six explicit reciprocal points gives 2,120 catalogues across ordinary/grey and scalar/spinful variants. Their 31,079 atomic columns contain 707,038 integer entries. All column ranks, antiunitary partner conditions and 217,553 directed column/segment comparisons agree in serial release, bounds-checked debug and two-rank MPI runs. These are sampled atomic reference matrices, not 2,120 separate material calculations or complete EBR catalogues.

The integer layer factors all 2,120 generated matrices and recovers signed test vectors and 33,199 nonnegative witnesses. Independent small-matrix tests compare the Smith invariants with

gcds of all minors and check unimodular transforms for 384 rectangular or rank-deficient matrices. All 6,642 two-Kramers-pair inversion patterns at four/eight TRIM agree with the separate analytic Fourier criterion; 18,225 additional nonnegative systems agree with exhaustive bounded coefficient enumeration. Arithmetic and search limits are tested as unresolved outcomes. This evidence concerns exact signature equations, not material topology or general EBR elementarity.

The Gaussian-band integration supplies signed and nonnegative atomic witnesses for the small He/Ne GPW, GAPW and SOC examples, including sheared cells and screw sectors. The single screw branch fails integer membership, consistently with its independent reciprocal-loop monodromy obstruction. A zero-budget TRIM test reports signed membership and unresolved nonnegative membership rather than mistaking an unfinished search for an obstruction. On all 2,120 generated reference matrices, reordered operations and irreps and integer coset shifts preserve the recovered signature after checking the independently regenerated factors. Corrupted factors and unmatched operations are rejected. The focused serial and two-rank MPI drivers each pass 135 assertions, combining six native unit programs with 129 Gaussian assertions, including automatic reciprocal sampling and the joint compatibility quotient. Another 36 export/k-point assertions pass in each configuration.

The reciprocal-geometry backend independently checks all 530 Hall settings and their grey extensions. It generates 18,700 fixed-family closures, 57,357 specialization incidences and 18,730 primitive torus cycles. Stabilizers agree with the full spinful little-group kernel; 333,900 rational-grid points have an unambiguous matching family. Coordinate changes preserve the families and incidence graph. The induced scalar/spinful atomic columns satisfy 2,481,572 directed incidence/cycle comparisons, including antiunitary restrictions when present. Serial, two-rank MPI and instrumented-kernel runs agree. These are geometric and representation tests, not additional DFT materials or a proof of complete global band connectivity.

The automatic Gaussian path is tested for He/GPW, all-electron He/GAPW, Ne/GTH-SOC and an equivalent sheared He cell. Each generates 15 fixed families, 56 points and 82 compatible directed segments. Replaying these points and shifts through explicit NNKP input gives selected eigenvalues within 2.1×10^{-13} eV and band characters within 2.2×10^{-14} ; all original segment restrictions and atomic integer witnesses agree exactly. Refining to 138 points and 164 segments retains compatibility. A deliberately selected single screw branch instead fails the band-isolation guard at a generated crossing. Invalid sample budgets and Wilson-mode combinations are also rejected.

The joint integer construction additionally certifies the compatibility kernel and atomic quotient for every ordinary/grey and scalar/spinful Hall-setting graph. Exact tests recover the four/eight-point inversion quotients $\mathbb{Z}_2$ and $\mathbb{Z}_2^3 \times \mathbb{Z}_4$ and compare quotient classes with direct signed membership for all 6,642 two-pair signatures. Independent arbitrary-precision Smith calculations validate the exported Gaussian certificates. The automatic scalar GPW/GAPW and sheared fixtures have compatible rank six and a single $\mathbb{Z}_2$ factor; the SOC fixture has rank five and factors $\mathbb{Z}_2^2 \times \mathbb{Z}_4$. Their occupied signatures have zero class. The single screw branch is incompatible and receives no class. Star identifications additionally include nonsymmorphic phases and antiunitary conjugation. In all 2,120 Hall-setting/symmetry combinations, both the compatible rank and finite quotient factors agree with the published 230-space-group tables [16]. This is an independent algebraic benchmark, not a proof of complete EBR labels or material topology. Partial explicit graphs can still retain free factors: the refined He Wilson example changes from 23 to four after star identification and remains incomplete.

Real-space specialization tests now independently check 22,592 induction relations in all 2,120 ordinary/grey scalar/spinful Hall variants. Eight reciprocal points per relation give 180,736 character identities; recursive composite expansions preserve the complete tested signatures and band dimensions. Analytic cases exercise lifted intermediate paths, origin/cell/coset conventions, antiunitary pairing, and malformed-witness rejection. The affected-space-group coverage agrees with the exception tables of Ref. [39]. An independent comparison with the Bilbao EBR data archived by Crystalline.jl [40] also matches the maximal-site generator count for every one of the

2,120 variants. These counts are not canonical-irrep-by-irrep matches. Optimized SSMP, each of two MPI ranks and an instrumented build give identical sweep records. Arbitrary-precision checks independently validate the exported integer expansions, lifted geometry and provenance in 15 Gaussian fixtures per build. The official drivers pass 171/171 assertions in both SSMP and MPI without changing physical reference values.

An independent affine-reference comparison goes beyond total counts: all 1,731 Wyckoff families of the 230 conventional space groups match, giving 6,924 correspondences across the four symmetry variants. Simultaneous basis shear, origin shift, integer coset changes and operation reversal preserve the named specialization graphs. At each individual Wyckoff position, the primitive-cell band-dimension multiset of the unreduced maximal-site generators agrees with the archived EBR table, covering 10,398 columns. Equal-dimensional onsite irreps are not distinguished by this dimension check; canonical identification of the complete onsite EBR columns requires additional character and convention information.

A separate character-level test compares all 2,700 scalar and spinful tables at the tabulated reciprocal points of the 230 space groups in `irreptables` 3.1.0 [21, 22]. All 8,907 irreps match bijectively, with largest residual 1.21×10^{-5} against the finite-precision reference data. The reference adapter explicitly uses the opposite tabulated Bloch wave-vector convention and the actual native spin lifts; it does not selectively conjugate characters or fit a phase independently for each irrep. The native algebra is checked at 10^{-9} , independently of the 2×10^{-4} tolerance for the rounded external tables. Optimized serial, both processes of a two-rank MPI run, and a focused instrumented executable give identical 11,607 normalized match and label records. This establishes named reference correspondences, not automatic standard labels in the Gaussian analysis or complete EBR identities and connectivity.

A separate onsite correspondence retains the Cartesian axes and the actual spin lifts of the site stabilizer. For 460 scalar/spinful Gamma reference tables, three coordinate variants yield 1,380 successful complete-table matches. Their 88,602 reference-normalized character entries agree across optimized serial, two MPI processes, and a focused instrumented build within 8.44×10^{-15} ; coordinate changes alter them by at most 9.77×10^{-15} . This test removes the space-group translations to realize onsite point groups. A complementary $P6_3$ two-site model verifies named C_3 onsite columns and their induced screw phases, including spin and grey extensions. These checks distinguish representation content from arbitrary internal column numbering, but do not yet identify every actual Wyckoff-site EBR or assign standard labels automatically to Gaussian bands.

Keeping the actual fractional translations, a subsequent test covers generic representatives of all 1,731 conventional Wyckoff families. Scalar/spinful ordinary/grey choices and two coordinate conventions give 13,848 complete onsite matches. An independent orbit sum then verifies 83,088 induced character tables at six reciprocal points. Their 1,384,328 character entries agree across builds and coordinate changes within 2.67×10^{-14} . This validates real-site induction and its Bloch phases in declared reference axes; it does not identify the full canonical EBR catalogue or prove global band connectivity.

Combining independently tabulated EBR vectors with these character conventions gives a stronger, site-resolved check of 5,641 named scalar/spinful reference columns in all 230 ordinary space groups. Every reference matches one consistent native onsite column over its entire tabulated reciprocal-point sequence, also after combined cell, origin, axis and coset changes. Of these columns, 5,263 have a unique candidate within their specified Wyckoff position; 378 remain indistinguishable at the sampled points. The latter are retained as candidate sets, not assigned unique onsite labels or declared equivalent Bloch bundles. A subsequent finite-Fourier comparison on the entire invariant reciprocal tori, including their disconnected components, finds equal unitary character functions for all 378 multi-candidate references. Thus finer reciprocal sampling alone cannot resolve these particular ambiguities; character equality still does not establish bundle equivalence. The largest dimension-scaled character residual is 1.05×10^{-5} against the rounded source tables. The SI records the independent opposite-valley references, negative tests and reproducible data.

This check does not replace canonical onsite conventions or global connectivity analysis.

A Gaussian post-SCF GTH-SOC test independently checks that the spin lifts used for the factor algebra and for the actual band sewing agree. For a Ne Kramers doublet in a primitive body-centered cubic cell, independent quaternion reconstruction gives a product-law residual of 2 despite passing the metric, subspace and energy tests. Reusing the stored lifts reduces this residual to 8.5×10^{-14} without altering the input or validation tolerances; the SCF energies agree within 3×10^{-14} hartree. The property point and the SCF point are both Gamma; the primitive-basis $2 \times 2 \times 2$ mesh is not a full-cubic-symmetry reference. Details and the separate quadrature convergence checks are given in the Supporting Information.

11.4 Convergence and material validation

This methods study assesses implementation correctness through independent reference calculations, operator identities, covariance tests and agreement between numerical backends and parallel configurations. Increasing the basis or system size supplies additional robustness tests, but convergence of a material observable is not itself a test of implementation correctness. The TZVP controls extend the tested basis and state spaces and reproduce the independent reference results at the specified settings. Systematic material-specific convergence of basis, size, sampling, temperature and dissipation belongs to subsequent application studies. The retained parameter scans document both resolved results and transitions, providing reproducible starting points for those studies.

12 Conclusions

A common nonorthogonal operator representation connects several otherwise distinct ways of analysing electronic states in CP2K. Metric-aware current operators support finite-temperature transport; directed Gaussian overlaps support Wilson and parameter-space invariants; and Hamiltonian-position tuples support local indices and joint approximate states. Sparse assembly and factorization address their numerical cost without replacing the physical assumptions of each construction.

The numerical validation combines independent operator reconstructions, dense and sparse solver comparisons, and serial and MPI calculations for model systems and GPW/GAPW/SOC electronic structures. The 96-atom Bi flake provides a larger finite-system benchmark with a complete 3,264-spinor TZVP space. All four native localizer controls reproduce the independently reconstructed indices, with gap differences below 3.02×10^{-11} hartree. Together with the neon and stanene band-topology demonstrations and the Si/Al k-point tests of the companion study, these results establish a quantitatively validated implementation for the methods and parameter ranges examined here.

The fixed-geometry DZVP/TZVP comparison also measures the physical basis dependence of the Bi flake. Eight of nine sampled indices agree, while the remaining change is reproduced by both numerical paths. The TZVP calculation therefore provides a validated reference for this finite system, with its basis sensitivity explicitly quantified. Bulk limits and complete-basis extrapolations are separate from these implementation tests and belong to subsequent material-specific studies. The TopoHSE-DB and generative-predictive discovery studies illustrate the broader materials-science applications motivating the infrastructure, rather than additional validation data generated in this work.

The crystalline framework now resolves projective unitary irreps, antiunitary pairing and doubling, and their restrictions along explicit reciprocal-space segments. Its algebraic coverage across conventional cell settings and supplied magnetic groups is distinct from a magnetic electronic-structure implementation or a complete EBR catalogue. The shared operator representation and reproducible validation records provide a basis for applying these complementary analyses to

further materials and for separating electronic-structure effects from numerical and finite-size effects.

Code and Data Availability

The manuscript sources, Supporting Information, and retained validation records are maintained in the development repository <https://github.com/DCM-Uni-Paderborn/TopologicalCP2K>. The repository currently has restricted access during manuscript preparation; public archival availability will be established before publication. CP2K is distributed under the GNU General Public License, version 2 or later, at <https://github.com/cp2k/cp2k>. The working implementation and retained calculation records are described in the Supporting Information. Not all developments described here are present in an upstream release. Source-reconstruction patches, build configurations, dependency fingerprints, and repeat calculations for the Wilson, localizer, and Pfaffian paths accompany the Supporting Information. These records distinguish new builds from reanalysis of existing matrices. A consolidated snapshot of the current tracked CP2K source is provided as a patch against a public upstream base. Reapplying it reproduces the exact source-tree identity without modifying the verification checkout. Historical calculations retain their own source versions and dependency fingerprints. External-library notices are retained separately, with unresolved licensing assignments explicitly identified. Public archival availability and the final licensing review must be completed before submission. The present draft does not assign an archival DOI or imply that the development branch has already been released.

Authorship and Declarations

The author list and corresponding author follow the agreed manuscript scope. Affiliations, CRediT roles, funding, competing-interest declarations, and any required disclosure concerning writing tools are to be confirmed by all authors before submission.

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